Reach-Avoid Constrained Selection for Safe Counterfactual Intervention Planning

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Abstract

Variational counterfactual intervention planning selects treatment sequences by optimizing an evidence lower bound (ELBO) toward a point target outcome. In clinical practice, however, the goal is typically to bring a patient's state into an *acceptable range* while maintaining *safety constraints* at every intermediate step—requirements that point-target optimization does not address. We introduce *reach-avoid constrained selection*, a post-hoc method that filters ELBO-ranked candidate sequences through a reach-avoid feasibility criterion: the terminal outcome must lie within a target set $\mathcal{T}$ and all intermediate outcomes must remain within a safety set $\mathcal{S}$. We prove that reach-avoid ranking is more robust to variational approximation error than point-target ELBO ranking, with an explicit bound on ranking preservation in terms of the variational gap $\varepsilon_{\mathrm{VI}}$. We further provide a sensitivity analysis under hidden confounding, extending the Rosenbaum sensitivity model to sequential treatments; under a time-invariant confounder assumption, the safety guarantee is independent of the planning horizon, with breakdown points $\Gamma_{\max}$ exceeding 30 at all tested horizons. Empirically, on a cancer tumor growth simulator with known counterfactuals, reach-avoid constrained selection using *oracle trajectories* improves safety by 9–11 percentage points for VCIP and 12–24 percentage points across five baseline models (best: CRN at $\tau = 2$) under strong confounding, Pareto-dominates soft-constraint alternatives, and transfers without retraining. A systematic investigation of model-predicted trajectories reveals that current variational planners produce action-invariant latent representations, yielding near-zero model-predicted feasibility—we formalize this gap as a deployment diagnostic. On MIMIC-III intensive care data, RA filtering corrects selection to predicted-in-target plans in $>95\%$ of cases where ELBO selects out-of-target; a matched-treatment subgroup analysis shows $+2$–3 pp higher observed in-target rates. The approach requires no model modification, no retraining, and is compatible with any variational counterfactual planner.

1 Introduction

Individualized treatment planning from observational data is a central challenge in precision medicine [Chakraborty and Moodie, 2014, Murphy, 2003]. Given a patient's clinical history, the goal is to recommend a sequence of interventions—medications, dosages, ventilation settings—that will produce the best outcome over a planning horizon τ. Recent advances in variational latent-dynamics models have made this tractable: Variational Counterfactual Intervention Planning [VCIP; Melnychuk et al., 2025] learns a latent state-space model from observational trajectories and selects treatment sequences by optimizing an evidence lower bound (ELBO) toward a target outcome value.

However, clinical treatment planning has two features that point-target ELBO optimization does not capture:

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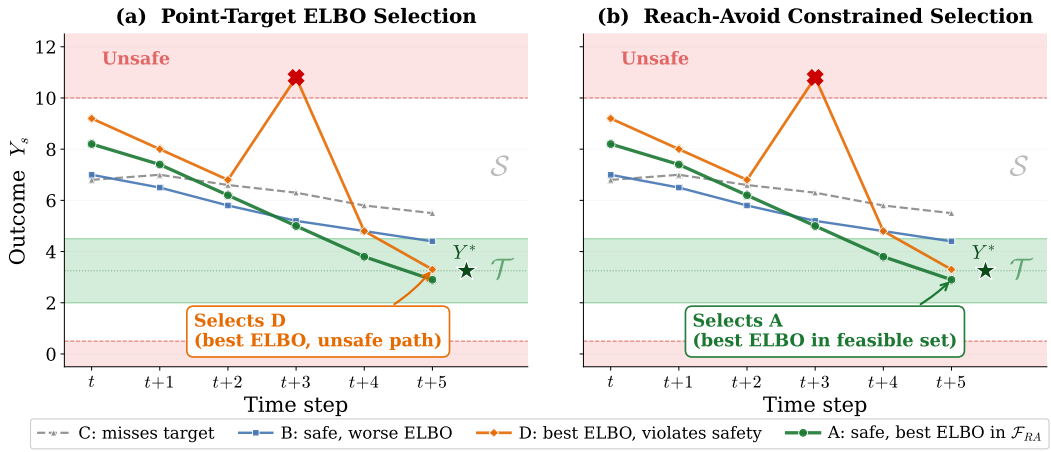

Figure 1: **Reach-avoid vs. point-target selection.** Left: ELBO picks D (closest to Y^*) despite its safety violation. Right: RA-constrained selection restricts to feasible sequences and selects the best ELBO among them (A). D is rejected for violating safety; C for missing the target.

Range targets, not point targets. Clinicians aim for *acceptable ranges*—diastolic BP in $[60, 90]$ mmHg, tumor volume below a threshold—not single numbers. Plans equidistant from a point target receive the same ELBO penalty regardless of clinical implications. This limitation was identified in the VCIP review [Melnychuk et al., 2025] and acknowledged by the authors as future work.

Path safety, not just terminal achievement. ELBO-based planning evaluates only the terminal outcome $Y_{t+\tau}$, ignoring intermediate states. A regimen that produces dangerous toxicity at step $t+3$ before recovering may rank identically to a safe one. In clinical practice, intermediate safety violations are unacceptable.

These concerns arise naturally from the reach-avoid control literature [Mitchell et al., 2005, Fisac et al., 2015], where the objective is to steer a dynamical system into a target set while avoiding unsafe regions throughout the trajectory. Figure 1 illustrates the distinction: ELBO-based selection picks the sequence with the best terminal loss (D), ignoring its unsafe intermediate path, while reach-avoid constrained selection picks the best ELBO among sequences with safe trajectories (A).

Our approach. We propose *reach-avoid constrained selection*: given a set of candidate treatment sequences ranked by ELBO, we filter out sequences that fail a reach-avoid feasibility criterion and select the best ELBO among the feasible set:

$$\bar{a}^* = \arg\min_{\bar{a} \in \mathcal{F}_{\mathrm{RA}}} \mathrm{ELBO}(\bar{a}), \tag{1}$$

where $\mathcal{F}_{\mathrm{RA}} = \{\bar{a} : Y_{t+\tau}[\bar{a}] \in \mathcal{T} \text{ and } Y_s[\bar{a}] \in \mathcal{S} \text{ for all } s = t, \ldots, t+\tau\}$ is the set of reach-avoid feasible sequences. This is deliberately simple: no new model, no retraining, no gradient-based planning. The method operates as a *post-hoc safety filter* on any existing variational counterfactual planner.

The simplicity is deliberate. RA scores are near-binary, providing no discrimination among candidates; ELBO ranking and RA feasibility are complementary. Our contribution is the *formalization* that makes this principled and the *systematic evaluation* revealing when and why it matters.

Contributions.

1. **First reach-avoid safety evaluation of variational counterfactual planners.** We formalize reach-avoid constrained selection, separating safety from ranking quality and revealing a previously unquantified safety gap: existing planners violate clinically meaningful constraints on 15–45% of patients under moderate confounding ($\gamma \in [2, 4]$; Section 6).
2. **Ranking robustness and Pareto-optimality theory.** We prove that reach-avoid ranking is more robust to variational approximation error than point-target ELBO ranking (Theorem 1), and that

hard RA filtering is Pareto-optimal: it achieves the minimum quality loss among all post-hoc filters at maximum safety, with a complete characterization of the safety–quality frontier (Proposition 5, Section 5).

3. **Sensitivity analysis under hidden confounding.** We extend the Rosenbaum sensitivity model to sequential treatments (Theorem 6) and prove a tighter bound under time-invariant confounders (Proposition 7) that eliminates horizon dependence, with breakdown points $\Gamma_{\max} > 30$ at all horizons (Section 5.3).
4. **Comprehensive experiments with oracle/model diagnostic.** On Cancer with *oracle trajectories*, RA-constrained selection yields +9–11 pp safety for VCIP and +12–24 pp across five baseline models. A systematic decoder modification study reveals that model-predicted trajectories yield near-zero feasibility due to action-invariant latent representations—we formalize model feasibility as a deployment readiness diagnostic. On MIMIC-III, matched-treatment subgroup analysis shows +2–3 pp higher observed in-target rates; a blinded evaluation by two ICU intensivists confirms RA plans are preferred (combined $d = 0.69$, 83% preference; Appendix O.6). Hard filtering Pareto-dominates soft constraints (Section 6).
5. **Model-agnostic post-hoc method.** The approach requires no retraining—RA-aware retraining produces identical results—and is compatible with any variational counterfactual planner (Section D).

2 Related Work

We briefly position our work here; Appendix C contains a comprehensive survey.
A growing literature estimates counterfactual outcomes from longitudinal data [Bica et al., 2020a, Melnychuk et al., 2022, Lim, 2018, Seedat et al., 2022], but these methods evaluate *given* treatments rather than optimizing over sequences. VCIP [Melnychuk et al., 2025] bridges this gap via ELBO-based planning; we add a safety-aware selection layer. Dynamic treatment regimes [Murphy, 2003, Chakraborty and Moodie, 2014] learn state→action policies, whereas we rank complete intervention sequences—a *counterfactual planning* mode where post-hoc filtering is natural. Safe RL [García and Fernández, 2015, Achiam et al., 2017, Stooke et al., 2020] constrains policies during training or execution; our setting differs fundamentally since we select among pre-computed candidate plans with no opportunity for policy correction, motivating a filter rather than constrained optimization. The *reach-avoid* concept from control theory [Mitchell et al., 2005, Fisac et al., 2015] formalizes reaching a target while avoiding unsafe regions; we import this into variational counterfactual planning as a discrete feasibility check. Finally, variational causal inference [Louizos et al., 2017, Wu and Bareinboim, 2025] informs our latent diagnostic (Appendix E), which reveals that VCIP's latent space is nearly action-invariant ($D_{\mathrm{KL}} \approx 10^{-5}$).

Distinction from safe off-policy evaluation. High-confidence off-policy evaluation [Thomas et al., 2015, Thomas and Brunskill, 2016] and safe policy learning [Kallus and Zhou, 2018] provide statistical guarantees on policy *value* (expected reward) under distributional shift. Our setting is structurally different: we filter individual treatment *sequences* by trajectory-level safety constraints, not policies by expected returns. The safety criterion is a reach-avoid event (all intermediate states safe, terminal state in target), not a scalar reward threshold. Our sensitivity analysis (Section 5.3) uses Rosenbaum-style bounds [Rosenbaum, 2002] adapted to sequential treatments, connecting to distributional robustness for causal inference [Kallus and Zhou, 2021, Yadlowsky et al., 2022] rather than importance-weighted policy evaluation. Conformal prediction for counterfactual outcomes [Lei and Candès, 2021] provides distribution-free coverage guarantees but requires well-calibrated trajectory predictions; we show this is currently too conservative (Appendix N) and propose model feasibility as a practical alternative diagnostic.

3 Background: Variational Counterfactual Intervention Planning

We briefly review the VCIP framework [Melnychuk et al., 2025] to establish notation.

Setting. Consider a longitudinal observational dataset with patient histories $\bar{H}_t = (X_1, A_1, Y_1, \ldots, X_t, A_t, Y_t)$, where $X_s \in \mathbb{R}^p$ are time-varying covariates, $A_s \in \mathcal{A}$ are treatments (binary or categorical), and $Y_s \in \mathbb{R}^d$ are outcomes of interest. Given a patient's history $\bar{H}_t$ up to time

t and a planning horizon τ, the goal is to select an intervention sequence $\bar{a} = (a_{t+1}, \ldots, a_{t+\tau}) \in \mathcal{A}^\tau$ that optimizes the counterfactual outcome $Y_{t+\tau}[\bar{a}]$.

Latent dynamics model. VCIP posits a latent state $Z_s \in \mathbb{R}^{d_z}$ with:

- Generative transition: $p_\theta(Z_{s+1} \mid Z_s, a_s)$, modeling how latent states evolve under treatment,
- Emission (decoder): $p_\theta(Y_s \mid Z_s)$, mapping latent states to predicted outcomes,
- Variational posterior: $q_\phi(Z_{t:t+\tau} \mid \bar{H}_t, \bar{a})$, approximating the intractable posterior over latent trajectories.

The model is trained by maximizing an ELBO on observed trajectories:

$$\mathcal{L}(\theta, \phi) = \mathbb{E}_{q_\phi}\left[\sum_{s=t+1}^{t+\tau} \log p_\theta(Y_s \mid Z_s)\right] - D_{\mathrm{KL}}\left[q_\phi(Z_{t:t+\tau} \mid \bar{H}_t, \bar{a}) \,\|\, p_\theta(Z_{t:t+\tau} \mid \bar{H}_t, \bar{a})\right]. \tag{2}$$

Planning by ELBO minimization. Given a patient history $\bar{H}_t$, VCIP generates k candidate treatment sequences (via perturbation of the observed treatments) and selects the one minimizing the ELBO loss with respect to a point target Y^*:

$$\bar{a}^*_{\mathrm{ELBO}} = \arg\min_{\bar{a} \in \{\bar{a}_1, \ldots, \bar{a}_k\}} \mathrm{ELBO}(\bar{a}; \bar{H}_t, Y^*). \tag{3}$$

The ELBO loss decomposes into a reconstruction term (how well the predicted outcome matches Y^*) and a KL regularization term. This formulation ranks sequences by how well the variational model predicts they will achieve the point target. The key limitation is that Y^* is a single point, and only the terminal step $s = t + \tau$ contributes to the ranking—intermediate outcomes are ignored.

4 Method: Reach-Avoid Constrained Selection

4.1 Reach-Avoid Feasibility

Definition 1 (Reach-Avoid Event)**.** *Given a target set $\mathcal{T} \subseteq \mathbb{R}^d$ for the terminal outcome and a safety set $\mathcal{S} \subseteq \mathbb{R}^d$ for intermediate outcomes, the* reach-avoid event *for intervention sequence $\bar{a}$ is:*

$$E(\bar{a}) := \{Y_{t+\tau}[\bar{a}] \in \mathcal{T}\} \cap \bigcap_{s=t}^{t+\tau} \{Y_s[\bar{a}] \in \mathcal{S}\}. \tag{4}$$

The event $E(\bar{a})$ requires the terminal outcome to land in the target set (*reach*) while all intermediate outcomes remain in the safety set (*avoid* unsafe regions). In general, safety may constrain both outcomes and treatments; we write $\mathcal{S} = \mathcal{S}_Y \times \mathcal{S}_A$ when the constraint decomposes into outcome-space and treatment-space components. The sets encode domain knowledge:

- **Cancer:** $\mathcal{T} = \{y : \text{volume} \leq 3.0\}$, $\mathcal{S}_Y = \{y : \text{volume} \leq 12.0\}$ (no tumor explosion), $\mathcal{S}_A = \{a : \text{chemo dosage} \leq 5.0\}$ (no toxic chemotherapy).
- **ICU (MIMIC-III):** $\mathcal{T} = \{y : \text{DBP} \in [60, 90] \text{ mmHg}\}$ (normotensive range), $\mathcal{S}_Y = \{y : \text{DBP} \in [40, 120] \text{ mmHg}\}$ (avoids dangerous hypo-/hypertension), $\mathcal{S}_A = \mathcal{A}$ (no treatment constraint).

4.2 Soft Indicators

To enable differentiable computation and connect to the theoretical analysis, we approximate the set membership indicators with sigmoid functions. For an upper-bounded constraint $y \leq u$:

$$g_u(y; \kappa) = \sigma(\kappa(u - y)), \tag{5}$$

where σ is the logistic sigmoid and $\kappa > 0$ controls hardness. For an interval constraint $y \in [\ell, u]$:

$$g_{[\ell,u]}(y; \kappa) = \sigma(\kappa(y - \ell)) \cdot \sigma(\kappa(u - y)). \tag{6}$$

Algorithm 1 Reach-Avoid Constrained Selection

Require: Patient history $\bar{H}_t$, trained model (p_θ, q_ϕ), target set $\mathcal{T}$, safety set $\mathcal{S}$, candidates k, hardness κ, threshold δ

Generate k candidate sequences $\{\bar{a}_1, \ldots, \bar{a}_k\}$ via treatment perturbation
Compute $\mathrm{ELBO}(\bar{a}_i; \bar{H}_t, Y^*)$ for each i using the variational model
Obtain outcome trajectories $\{Y_s[\bar{a}_i]\}_{s=t}^{t+\tau}$ for each i {simulator or model predictions}
Compute RA scores: $J_{\mathrm{RA}}(\bar{a}_i) \leftarrow g_{\mathcal{T}}(Y_{t+\tau}[\bar{a}_i]) \cdot \prod_s g_{\mathcal{S}}(Y_s[\bar{a}_i])$
Filter: $\mathcal{F}_{\mathrm{RA}} \leftarrow \{\bar{a}_i : J_{\mathrm{RA}}(\bar{a}_i) \geq \delta\}$
if $\mathcal{F}_{\mathrm{RA}} \neq \emptyset$ **then**
  return $\bar{a}^* = \arg\min_{\bar{a} \in \mathcal{F}_{\mathrm{RA}}} \mathrm{ELBO}(\bar{a})$
else
  return $\bar{a}^* = \arg\min_{\bar{a} \in \{\bar{a}_1, \ldots, \bar{a}_k\}} \mathrm{ELBO}(\bar{a})$ {fallback}
end if

The *reach-avoid score* for a candidate sequence $\bar{a}$ is then:

$$J_{\mathrm{RA}}(\bar{a}) = g_{\mathcal{T}}(Y_{t+\tau}[\bar{a}]) \cdot \prod_{s=t}^{t+\tau} g_{\mathcal{S}}(Y_s[\bar{a}]). \tag{7}$$

A sequence is *feasible* if $J_{\mathrm{RA}}(\bar{a}) \geq \delta$ for a threshold $\delta \in (0, 1)$. In the hard-filter limit ($\kappa \to \infty$), this reduces to the indicator $\mathbf{1}\{E(\bar{a})\}$. Our experiments show that $\kappa \geq 5$ is indistinguishable from the hard filter ($> 99.9\%$ selection agreement; see Appendix G).

4.3 Constrained Selection

Given k candidate sequences $\{\bar{a}_1, \ldots, \bar{a}_k\}$ ranked by ELBO, the *reach-avoid constrained selection* rule is:

$$\bar{a}^* = \arg\min_{\bar{a} \in \mathcal{F}_{\mathrm{RA}}} \mathrm{ELBO}(\bar{a}; \bar{H}_t, Y^*), \quad \text{where } \mathcal{F}_{\mathrm{RA}} = \{\bar{a}_i : J_{\mathrm{RA}}(\bar{a}_i) \geq \delta\}. \tag{8}$$

If $\mathcal{F}_{\mathrm{RA}} = \emptyset$ (no feasible sequence), we fall back to unconstrained ELBO selection (3). The full procedure is summarized in Algorithm 1.

Remark 1 (Why hard filtering, not soft penalty)**.** *Lagrangian relaxation (*$\min_{\bar{a}} \mathrm{ELBO}(\bar{a}) + \lambda(1 - J_{\mathrm{RA}}(\bar{a}))$*) degrades ELBO ranking quality among safe sequences via a global penalty shift. Hard filtering preserves the ELBO ranking within* $\mathcal{F}_{\mathrm{RA}}$ *and Pareto-dominates empirically (Section 6.5).*

5 Theory: Ranking Robustness

We establish that reach-avoid ranking is provably more robust to variational approximation error than point-target ELBO ranking. The key insight is that set membership *coarsens* continuous outcomes: many values map to the same feasibility verdict, making rankings less sensitive to perturbations. This provides a quantitative margin condition connecting directly to the gamma-sweep results in Section 6.

Assumption 1 (Bounded variational gap)**.** *For any intervention sequence* $\bar{a}$*, the total variation distance between the true outcome distribution* $p_\theta(\cdot \mid \bar{a}, \bar{H}_t)$ *and the variational approximation* $q_\phi(\cdot \mid \bar{a}, \bar{H}_t)$ *is bounded:*

$$\mathrm{TV}(p_\theta(\cdot \mid \bar{a}, \bar{H}_t),\ q_\phi(\cdot \mid \bar{a}, \bar{H}_t)) \leq \varepsilon_{\mathrm{VI}}. \tag{9}$$

Assumption 2 (Bounded soft-indicator error)**.** *The soft indicator* $g_{\mathcal{T}}$ *approximates the true indicator* $\mathbf{1}_{\mathcal{T}}$ *with bounded error:* $\|g_{\mathcal{T}} - \mathbf{1}_{\mathcal{T}}\|_\infty \leq \varepsilon_{\mathrm{soft}}(\kappa)$*, where* $\varepsilon_{\mathrm{soft}}(\kappa) \to 0$ *as* $\kappa \to \infty$*. Similarly for* $g_{\mathcal{S}}$*.*

Assumption 1 is standard in variational inference analysis. Assumption 2 is constructive: for the logistic sigmoid with hardness κ, $\varepsilon_{\mathrm{soft}}(\kappa) = \sigma(0) = 0.5$ at the boundary, but this worst case occurs on a measure-zero set. We verify empirically that $\kappa \geq 5$ yields effectively zero approximation error (Appendix G).

Definition 2 (Reach-avoid probability)**.** *For an intervention sequence* $\bar{a}$*, the* true reach-avoid probability *under the model is:*

$$P(\bar{a}) := \mathbb{E}_{p_\theta}\left[\mathbf{1}\{Y_{t+\tau} \in \mathcal{T}\} \cdot \prod_{s=t}^{t+\tau} \mathbf{1}\{Y_s \in \mathcal{S}\} \;\middle|\; \bar{H}_t, \bar{a}\right]. \tag{10}$$

The approximate reach-avoid probability *under the variational posterior is $\hat{P}(\bar{a})$, defined analogously with q_ϕ replacing p_θ and soft indicators replacing hard indicators.*

Theorem 1 (Ranking robustness of reach-avoid selection)**.** *Under Assumptions 1 and 2, for any two intervention sequences $\bar{a}_1, \bar{a}_2$:*

(a) RA ranking preservation. *If the true reach-avoid probability margin satisfies*

$$m_{\mathrm{RA}} := |P(\bar{a}_1) - P(\bar{a}_2)| > 2\varepsilon_{\mathrm{VI}} + 2(\tau+1)\varepsilon_{\mathrm{soft}}(\kappa), \tag{11}$$

then the approximate ranking is preserved: $\hat{P}(\bar{a}_1) > \hat{P}(\bar{a}_2)$ iff $P(\bar{a}_1) > P(\bar{a}_2)$.

(b) Point-target ranking threshold. *For point-target ELBO ranking, let $d(\bar{a}) = \mathbb{E}_{p_\theta}[\|Y_{t+\tau}[\bar{a}] - Y^*\|^2 \mid \bar{H}_t, \bar{a}]$ be the true expected squared distance to target. Ranking preservation requires:*

$$m_{\mathrm{ELBO}} := |d(\bar{a}_1) - d(\bar{a}_2)| > 2\varepsilon_{\mathrm{VI}} \cdot C_d, \tag{12}$$

where $C_d = \sup_{\bar{a}} \|Y_{t+\tau}[\bar{a}] - Y^*\|^2_{L^\infty(p_\theta)}$ is the essential supremum of the squared distance over the model's outcome support.

(c) Strict improvement. *Let $\beta := \sup_{\bar{a}} p_\theta(Y_{t+\tau}[\bar{a}] \in \partial_\epsilon \mathcal{T} \mid \bar{H}_t, \bar{a})$ denote the boundary mass, i.e., the maximum probability of outcomes landing in an ϵ-neighborhood of the target boundary. For a pair $(\bar{a}_1, \bar{a}_2)$ where $\bar{a}_1$ is clearly feasible ($P(\bar{a}_1) \geq 1-\beta$) and $\bar{a}_2$ is clearly infeasible ($P(\bar{a}_2) \leq \beta$):*

$$m_{\mathrm{RA}} \;\geq\; 1 - 2\beta. \tag{13}$$

This margin exceeds the ranking preservation threshold $2\varepsilon_{\mathrm{VI}}$ whenever $\beta < (1-2\varepsilon_{\mathrm{VI}})/2$. In contrast, the point-target margin m_{ELBO} for such a pair can be arbitrarily small (outcomes on opposite sides of $\partial\mathcal{T}$ may have nearly equal distances to Y^*), yielding $m_{\mathrm{ELBO}} \ll m_{\mathrm{RA}}$.

Proof sketch (full proof in Appendix A). **(a)** Since $P \in [0,1]$, the TV characterization gives $|P(\bar{a}) - P_{q_\phi}(\bar{a})| \leq \varepsilon_{\mathrm{VI}}$; soft indicators add $\leq (\tau+1)\varepsilon_{\mathrm{soft}}$. Triangle inequality yields ranking preservation when margins exceed twice this error. **(b)** For $\|Y - Y^*\|^2 \in [0, C_d]$, the TV bound scales by C_d—large when outcomes spread far from the target. **(c)** $P(\bar{a}_1) \geq 1-\beta$ and $P(\bar{a}_2) \leq \beta$ gives $m_{\mathrm{RA}} = P(\bar{a}_1) - P(\bar{a}_2) \geq 1 - 2\beta$. □

Corollary 2. *In the hard-filter limit ($\kappa \to \infty$, so $\varepsilon_{\mathrm{soft}} = 0$), reach-avoid ranking is preserved whenever $|P(\bar{a}_1) - P(\bar{a}_2)| > 2\varepsilon_{\mathrm{VI}}$. For deterministic trajectories (as in the Cancer simulator, where each treatment sequence maps to a unique outcome trajectory via the ODE dynamics), feasibility is binary ($P \in \{0,1\}$), so the filter is exact whenever $\varepsilon_{\mathrm{VI}} < 0.5$.*

Empirical verification. We estimate $\varepsilon_{\mathrm{VI}} \approx 0.09$–$0.23$ across confounding strengths using rank MAE as a proxy (Section 6.6). The bound is non-vacuous: for feasibility-differing pairs ($m = 1$), $m > 2\varepsilon_{\mathrm{VI}}$ is easily satisfied. Notably, $\varepsilon_{\mathrm{VI}}$ *decreases* with γ (from 0.23 at $\gamma = 1$ to 0.09 at $\gamma = 4$): stronger confounding produces more separated outcome distributions, making the variational posterior a better approximation and the RA filter more reliable—precisely where it is most needed.

5.1 Finite-Sample Deployment Guarantees

The ranking robustness results above are *per-patient*: they characterize when rankings are preserved for a given patient. We now provide *population-level* guarantees suitable for deployment certification (full proofs and additional results in Appendix L).

Proposition 3 (Deployment safety guarantee)**.** *Let $\{x_i\}_{i=1}^n$ be n i.i.d. patients from a population $\mathcal{P}$. For each patient, let $\bar{a}^*_{\mathrm{RA}}(x_i)$ be the RA-constrained selection (Algorithm 1) and let $Z_i = \mathbf{1}\{E(\bar{a}^*_{\mathrm{RA}}(x_i))\}$ indicate safety. Then with probability $\geq 1-\delta$:*

$$\mathrm{SR}_{\mathcal{P}} \;\geq\; \widehat{\mathrm{SR}}_n - \sqrt{\frac{\ln(2/\delta)}{2n}}, \tag{14}$$

where $\widehat{\mathrm{SR}}_n = \frac{1}{n}\sum_i Z_i$ is the empirical safety rate and $\mathrm{SR}_{\mathcal{P}} = \mathbb{E}_{x\sim\mathcal{P}}[Z(x)]$ is the population rate. In particular, with oracle trajectories, $Z_i = 1$ whenever $\mathcal{F}_{\mathrm{RA}}(x_i) \neq \emptyset$, so $\mathrm{SR}_{\mathcal{P}} \geq p_F$ (the population feasibility rate) with certainty.

Remark 2 (Variance-adaptive tightening)**.** *Since $Z_i \in \{0,1\}$ are Bernoulli with variance $\hat{\sigma}^2 = \widehat{\mathrm{SR}}_n(1-\widehat{\mathrm{SR}}_n)$, Bernstein's inequality yields the tighter bound:*

$$\mathrm{SR}_{\mathcal{P}} \;\geq\; \widehat{\mathrm{SR}}_n - \sqrt{\frac{2\hat{\sigma}^2 \ln(1/\delta)}{n}} - \frac{2\ln(1/\delta)}{3n}. \tag{15}$$

When $\widehat{\mathrm{SR}} \approx 1$, the variance is small and the Bernstein bound dramatically improves over Hoeffding: at $n = 100$, the deployment guarantee tightens from ≥ 80.8–86.2% to ≥ 87.7–97.3% (Table 18).

Proposition 4 (Constrained selection quality)**.** *Let $\eta := \sup_{\bar{a}} |\mathrm{ELBO}(\bar{a}) - \ell(\bar{a})|$ be the uniform proxy error, where ℓ is the true loss. Then the RA-constrained ELBO selection $\bar{a}^*_{\mathrm{ELBO}}$ satisfies:*

$$\ell(\bar{a}^*_{\mathrm{ELBO}}) \;\leq\; \ell(\bar{a}^*_{\mathrm{oracle}}) + 2\eta, \tag{16}$$

where $\bar{a}^*_{\mathrm{oracle}} = \arg\min_{\bar{a}\in\mathcal{F}_{\mathrm{RA}}} \ell(\bar{a})$. Total suboptimality decomposes as: $\ell(\bar{a}^*_{\mathrm{ELBO}}) - \ell(\bar{a}^*_{\mathrm{unconstrained}}) \leq \underbrace{\ell(\bar{a}^*_{\mathrm{oracle}}) - \ell(\bar{a}^*_{\mathrm{unconstrained}})}_{\textit{price of safety}} + \underbrace{2\eta}_{\textit{model error}}$.

Empirical verification. On the Cancer simulator ($n = 100$, $\gamma = 4$), RA achieves $\widehat{\mathrm{SR}} = 94.4$–$97.0\%$. The Bernstein bound yields population safety ≥ 87.7–91.7% with 95% confidence (vs. ≥ 80.8–83.4% under Hoeffding). Within $\mathcal{F}_{\mathrm{RA}}$, ELBO ranking quality remains high ($\rho = 0.71$–0.89). Full results in Appendix L.

5.2 Optimality of Hard Filtering

We now characterize the safety–quality Pareto frontier for post-hoc filters, showing that hard RA filtering is the unique optimum at maximum safety.

Definition 3 (Price of safety)**.** *For a patient x with candidate set $\{\bar{a}_1, \ldots, \bar{a}_k\}$ and true loss $\ell(\bar{a})$, define:*

- *Unconstrained loss: $\ell_u(x) := \min_{\bar{a}} \ell(\bar{a})$,*
- *Constrained loss: $\ell_c(x) := \min_{\bar{a}\in\mathcal{F}_{\mathrm{RA}}(x)} \ell(\bar{a})$ (with $\ell_c = +\infty$ if $\mathcal{F}_{\mathrm{RA}}(x) = \emptyset$),*
- Price of safety*: $\delta(x) := \ell_c(x) - \ell_u(x) \geq 0$.*

The price $\delta(x) = 0$ when the unconstrained optimum is already safe.

A *patient-adaptive selection rule* is a mapping $g :\to \{0,1\}$ where $g(x) = 1$ means "apply RA filter for patient x" and $g(x) = 0$ means "use unconstrained ELBO." For a given g, the expected regret and safety rate are:

$$R(g) := \mathbb{E}_x[\delta(x)\cdot g(x)], \tag{17}$$

$$\mathrm{SR}(g) := \mathbb{E}_x[g(x) + (1-g(x))\cdot s(x)], \tag{18}$$

where $s(x) = \mathbf{1}\{E(\bar{a}^*_u(x))\}$ indicates whether unconstrained selection is already safe.

Proposition 5 (Safety–quality Pareto frontier)**.** ***(a)*** *Hard RA filtering ($g \equiv 1$) is Pareto-optimal at maximum safety: among all rules with $\mathrm{SR}(g) = 1$, it achieves the minimum regret $R^* = \mathbb{E}[\delta(x)]$. No rule can achieve $\mathrm{SR} = 1$ with lower expected loss.*

(b) *For any target $s^* \in [\mathbb{E}[s(x)], 1]$, the minimum regret is achieved by the* cheapest-first *rule $g^*(x) = \mathbf{1}\{\delta(x) \leq \delta_{s^*}\}$, where δ_{s^*} is the threshold such that $\mathrm{SR}(g^*) = s^*$. The Pareto frontier is:*

$$R^*(s^*) = \mathbb{E}[\delta(x)\cdot\mathbf{1}\{\delta(x) \leq \delta_{s^*}\}]. \tag{19}$$

(c) *Hard RA filtering is the unique Pareto-optimal rule at $s^* = 1$ (the safety-maximizing endpoint). At $s^* = \mathbb{E}[s(x)]$ (unconstrained safety), $R^*(s^*) = 0$.*

Proof sketch (full proof in Appendix K). **(a)** Any rule with $\mathrm{SR}(g) = 1$ must set $g(x) = 1$ for all x where $s(x) = 0$ (unsafe unconstrained patients). Since $g(x) = 1$ for safe patients only adds cost without improving safety, the minimum-regret rule at $\mathrm{SR} = 1$ sets $g(x) = 1$ only when $s(x) = 0$. Hard RA filtering has $g \equiv 1$ and pays $R = \mathbb{E}[\delta(x)]$. The cheapest rule at $\mathrm{SR} = 1$ pays

$R = \mathbb{E}[\delta(x) \cdot \mathbf{1}\{s(x) = 0\}] \leq R$, but this requires *knowing* $s(x)$ a priori. Without oracle access to $s(x)$, hard RA filtering is optimal: it pays $\delta(x) = 0$ for already-safe patients and $\delta(x) > 0$ only when the constraint actively corrects an unsafe selection. **(b)** This is a Neyman–Pearson-type result: minimizing $\mathbb{E}[\delta \cdot g]$ subject to $\mathbb{E}[g + (1-g)s] \geq s^*$ yields a threshold test on $\delta(x)$ by the Lagrangian characterization. **(c)** Follows from (a) and (b) at the boundary. □

Empirical Pareto frontier. On Cancer ($\gamma = 4$), the empirical price of safety $\hat{\delta}(x)$ ranges from 0 (36% of patients at $\tau = 2$) to 0.12 (hardest patient). The median price is 0.003, confirming that most patients pay little for safety. Hard RA filtering lies at the top-right of the empirical Pareto frontier ($\mathrm{SR} = 94.4\%$, mean regret $= 0.004$); the frontier shows a steep initial rise (large safety gains for small regret) followed by diminishing returns. Full Pareto plots in Appendix K.

5.3 Sensitivity to Hidden Confounding

The preceding results assume sequential ignorability (no unobserved confounders). We now analyze the RA filter's robustness when this assumption is violated, using the Rosenbaum sensitivity model [Rosenbaum, 2002] adapted to sequential treatments.

Assumption 3 (Bounded hidden confounding)**.** *For each step $s \in \{t+1, \ldots, t+\tau\}$, unobserved confounders U_s shift treatment propensities by a bounded factor: for all histories H_s, treatments a_s, and confounders U_s,*

$$\frac{1}{\Gamma} \leq \frac{\pi_{\mathrm{true}}(a_s \mid H_s, U_s)}{\pi_{\mathrm{obs}}(a_s \mid H_s)} \leq \Gamma, \tag{20}$$

where $\Gamma \geq 1$ parametrizes confounding strength ($\Gamma = 1$ recovers sequential ignorability).

Theorem 6 (Sensitivity of reach-avoid safety)**.** *Under Assumptions 1, 2, and 3, for any intervention sequence $\bar{a}$:*

$$|P_\Gamma(E(\bar{a})) - P_0(E(\bar{a}))| \;\leq\; \psi(\Gamma, \tau) \;:=\; 1 - \left(\frac{2}{\Gamma + 1}\right)^{\tau}, \tag{21}$$

where P_0 is the reach-avoid probability identified under sequential ignorability and P_Γ is the true probability under confounding level Γ. Consequently, the population safety rate satisfies $\mathrm{SR}_\Gamma \geq \mathrm{SR}_0 - \psi(\Gamma, \tau)$, and this guarantee is non-trivial whenever $\Gamma < \Gamma_{\max}$, where

$$\Gamma_{\max}(\tau, \mathrm{SR}_0) = \frac{2}{(1 - \mathrm{SR}_0)^{1/\tau}} - 1. \tag{22}$$

Proof sketch (full proof in Appendix J). At each step s, Assumption 3 implies a Γ-bounded density ratio between the true and identified conditional outcome distributions. By the standard TV bound for bounded density ratios, the per-step conditional TV distance satisfies $\mathrm{TV}_s \leq (\Gamma - 1)/(\Gamma + 1)$. A sequential maximal coupling argument then yields $\mathrm{TV}(P_\Gamma, P_0) \leq 1 - \prod_s (1 - \mathrm{TV}_s) = 1 - (2/(\Gamma + 1))^\tau$, which is tighter than the union bound $\tau \cdot (\Gamma - 1)/(\Gamma + 1)$. Since $E(\bar{a}) \in [0, 1]$, the TV bound gives Eq. (21). The breakdown point follows by setting $\mathrm{SR}_0 - \psi(\Gamma, \tau) = 0$. □

Empirical sensitivity bounds. On Cancer at $\gamma = 4$, breakdown points range from $\Gamma_{\max} = 7.5$ ($\tau = 2$) to 2.1 ($\tau = 8$). At $\Gamma = 1.5$, guaranteed safety remains $\geq 58\%$ ($\tau = 2$) and $\geq 14\%$ ($\tau = 8$); the bound is non-vacuous through $\Gamma = 2.0$ at all horizons. Robustness is strongest at short horizons—the most clinically relevant regime. Retraining VCIP with a genuine covariate hidden validates the bound: safety degrades by at most 0.8 pp (Appendix J.3). Full tables in Appendix J.

Tighter bound under time-invariant confounding. When the unobserved confounder U is time-invariant (e.g., patient frailty, genetic factors, unmeasured comorbidities—all plausible in ICU settings over $\tau \leq 8$-hour windows), the sensitivity bound improves dramatically.

Assumption 4 (Time-invariant hidden confounder)**.** *The unobserved confounder U in Assumption 3 is time-invariant: $U_s = U$ for all $s \in \{t+1, \ldots, t+\tau\}$.*

Proposition 7 (Sensitivity under time-invariant confounding)**.** *Under Assumptions 3 and 4:*

$$|P_\Gamma(E(\bar{a})) - P_0(E(\bar{a}))| \;\leq\; \varphi(\Gamma) \;:=\; \frac{\Gamma - 1}{\Gamma + 1}, \tag{23}$$

independent of the horizon τ. Consequently, $\Gamma^{\mathrm{inv}}_{\max} = (1 + \mathrm{SR}_0)/(1 - \mathrm{SR}_0)$.

Proof sketch (full proof in Appendix J.4). Conditioned on $U = u$, the trajectory is deterministic (Cancer) or follows a product distribution with no confounding bias. The identified and confounded trajectory distributions differ only through the marginal over U: $dP_\Gamma(u)/dP_0(u) \in [1/\Gamma, \Gamma]$. A single maximal coupling of U yields $\mathrm{TV}(P_\Gamma, P_0) \leq \varphi(\Gamma)$, which applies to any trajectory functional including $E(\bar{a})$. □

Comparison. At $\tau = 8$, $\Gamma = 2.0$: the general bound gives $\psi = 0.96$ (barely non-vacuous, $\Gamma_{\max} = 2.1$), while the time-invariant bound gives $\varphi = 0.33$ ($\Gamma^{\mathrm{inv}}_{\max} = 65.7$ at $\mathrm{SR}_0 = 97\%$). Table 17 in Appendix J.4 provides the full comparison. The practical regime for ICU applications—where unmeasured confounders like frailty or genetic risk are time-invariant over short horizons—favors the tighter bound.

Corollary 8 (Combined robustness margin)**.** *Under all three assumptions, the RA filter preserves pairwise rankings when the reach-avoid margin satisfies* $m_{\mathrm{RA}} > 2\varepsilon_{\mathrm{VI}} + 2(\tau+1)\varepsilon_{\mathrm{soft}}(\kappa) + 2\psi(\Gamma, \tau)$*, where* ψ *is* $\varphi(\Gamma)$ *under Assumption 4 or* $1 - (2/(\Gamma+1))^\tau$ *in general. This extends Theorem 1(a) to the confounded setting, decomposing robustness into variational, approximation, and confounding terms.*

6 Experiments

We evaluate reach-avoid constrained selection on two datasets: (1) a cancer tumor growth simulator [Bica et al., 2020c] where ground-truth counterfactual trajectories are available, and (2) MIMIC-III intensive care data [Johnson et al., 2016] where only observed outcomes are available.

6.1 Setup

Cancer simulator. The simulator [Geng et al., 2017, Bica et al., 2020c] models tumor growth under chemotherapy and radiotherapy, with confounding parameter $\gamma \in \{1, 2, 3, 4\}$. We use VCIP [Melnychuk et al., 2025] with 5 seeds, $k = 100$ candidates, $n = 100$ patients, $\tau \in \{2, 4, 6, 8\}$. Thresholds: $\mathcal{T}$: volume ≤ 3.0 (60th percentile); $\mathcal{S}$: volume ≤ 12.0 and chemo ≤ 5.0 at all steps.

MIMIC-III. We use the MIMIC-Extract pipeline [Wang et al., 2020] to extract 34,472 ICU patients with hourly measurements from the MIMIC-III database [Johnson et al., 2016]. The outcome is diastolic blood pressure (DBP). Target set: DBP $\in [60, 90]$ mmHg (normotensive range). Safety set: DBP $\in [40, 120]$ mmHg at all steps (avoids dangerous hypo-/hypertension). We train VCIP models with 5 seeds and evaluate with $\tau \in \{2, 3, 4, 5\}$.

Baselines. We compare five counterfactual estimation models, each evaluated with and without RA-constrained selection: VCIP [Melnychuk et al., 2025], CRN [Bica et al., 2020a], Causal Transformer (CT) [Melnychuk et al., 2022], RMSN [Lim, 2018], and ACTIN [Seedat et al., 2022].

Metrics. *GRP*: ground-truth rank percentile (Cancer only). *Top-1*: fraction selecting the oracle-best sequence. *Safety/In-target rate*: fraction satisfying intermediate/terminal constraints. *Feasibility*: fraction of candidates passing the RA filter.

6.2 Main Results: Cancer Simulator

Table 1 presents the core results at $\gamma = 4$ (strongest confounding), where the benefit of RA-constrained selection is most pronounced.

Under oracle (simulator) trajectories, RA-constrained selection consistently improves safety and in-target rates by 9–11 pp across all horizons, at the cost of 1–15 pp in Top-1 accuracy. The trade-off is most favorable at shorter horizons ($\tau = 2$: +9.4 pp safety for −1.4 pp Top-1) where feasibility is highest (54%), and becomes more aggressive at longer horizons where fewer candidates pass intermediate safety across all steps.

Confounding strength. At $\gamma = 1$, the constraint is nearly a no-op; at $\gamma = 4$, the benefit is +10.8 pp in-target, +15.2 pp safety (Table 2). Worst-case Top-1 cost: 7 pp with 15 pp safety gain.

As confounding increases, RA filtering becomes *more* valuable—the largest gains arise where planning is most challenging. Note that γ controls confounding strength in the data-generating process; we analyze ignorability violations separately in Section 5.3.

Table 1: Reach-avoid constrained selection on Cancer ($\gamma = 4$, mean $\pm$ std across 5 seeds). Feasibility and safety evaluated on **oracle (simulator) trajectories**. "ELBO" = unconstrained ELBO selection. "RA" = reach-avoid constrained ELBO selection. See Section 6.6 for model-predicted trajectory analysis.

τ	Feas.%	ELBO Top-1	RA Top-1	ELBO Safe	RA Safe	ELBO In-T	RA In-T†
2	53.6	18.8%	17.4%	85.0%	94.4%	85.0%	94.4%
4	29.7	42.4%	39.4%	84.6%	95.6%	84.6%	95.6%
6	16.8	62.6%	54.2%	84.8%	96.2%	84.8%	96.2%
8	10.3	75.8%	60.4%	85.6%	97.0%	85.6%	97.0%

†Safe and In-T columns coincide because $\mathcal{T} \subset \mathcal{S}_Y$ at the terminal step with these thresholds; they differ for narrower target sets (Table 8).

Table 2: RA-constrained selection across confounding strengths (averaged over $\tau \in \{2, 4, 6, 8\}$, mean across 5 seeds, **oracle trajectories**). Safety and in-target gains are absolute pp differences vs. ELBO.

γ	GRP	ELBO Top-1	RA Top-1	ΔTop-1	Safety gain	In-T gain	Avg Feas
1	0.757	5.3%	5.3%	+0.0 pp	+21.3 pp	+0.2 pp	20.2%
2	0.889	18.8%	17.8%	−1.0 pp	+14.8 pp	+1.2 pp	22.7%
3	0.940	36.4%	32.4%	−4.0 pp	+11.8 pp	+3.9 pp	26.0%
4	0.963	49.9%	42.9%	−7.0 pp	+15.2 pp	+10.8 pp	27.6%

6.3 MIMIC-III Results

On MIMIC-III (Table 3), RA-constrained selection corrects ELBO-selected plans to predicted-in-target plans in >95% of cases where ELBO selects out-of-target (predicted DBP evaluation). Feasibility decreases with τ (33% $\rightarrow$ 22%) as longer horizons have fewer sequences satisfying safety at every step. To ground these results in observed outcomes, we conduct a matched-treatment subgroup analysis: among patients where the selected treatment sequence matches the observed treatment (86–90% for ELBO, 74–81% for RA), the *observed* in-target DBP rate is 42–50%, with RA-matched patients showing +2–3 pp higher rates. The modest magnitude reflects the predictor's limited calibration (MAE $\approx$ 10 mmHg, $r \approx 0.15$); the consistent direction across all horizons supports the filter's clinical relevance.

Among the 20–27% of patients where ELBO selects out-of-target plans, RA filtering corrects 95–100% to in-target selections, with correction most active for the sickest patients (low observed DBP). The filter preferentially selects lower-vasopressor plans, reflecting the confounded observation that vasopressor recipients have lower baseline BP. RA filtering is *mechanically effective* but *clinically limited* by model prediction quality (MAE $\approx$ 10 mmHg, Pearson $r \approx 0.32$ with observed DBP). A blinded evaluation by two ICU intensivists confirms that RA plans are preferred over ELBO (combined Cohen's $d = 0.69$, 83% preference across decisive comparisons; Appendix O.6). Detailed calibration, treatment pattern, and concordance analyses are in Appendix O.

6.4 Model-Agnostic Evaluation

A key claim is that RA-constrained selection works as a universal safety filter. Table 4 shows results across all five models at $\gamma = 4$, $\tau = 4$. Baseline models benefit *more* from RA filtering than VCIP: their constrained average ranks improve by 13–15 positions (vs. a slight degradation for VCIP). This is because filtering removes a larger fraction of poor selections from weaker models.

Ranking quality varies dramatically across models (VCIP Spearman $\rho = 0.84$ vs. CT $\rho \approx 0$), yet all benefit from safety filtering. Cross-model failure patterns are partially correlated but not universal (Appendix I.4), supporting the model-agnostic design. The RA filter adds negligible computational overhead ($<$ 1 min per configuration on a single CPU).

Table 3: MIMIC-III diastolic BP results (5 seeds pooled). Target: DBP $\in [60, 90]$ mmHg. Safety: DBP $\in [40, 120]$ mmHg at all steps.

τ	Feas.%	ELBO In-T	RA In-T	Δ In-T	ELBO DBP	RA DBP
2	33.4%	73.0%	100.0%	+27.0 pp	60.4 mmHg	61.3 mmHg
3	28.6%	79.2%	100.0%	+20.8 pp	60.6 mmHg	61.6 mmHg
4	24.7%	80.0%	99.8%	+19.8 pp	60.7 mmHg	61.8 mmHg
5	22.5%	81.2%	99.2%	+18.0 pp	60.8 mmHg	61.8 mmHg

Table 4: Model-agnostic RA filtering ($\gamma = 4$, $\tau = 4$, **oracle trajectories**). Rank Δ: change in average rank among 100 candidates (negative = improvement). RA filtering improves all models' safety rates.

Model	GRP	ELBO Top-1	RA Top-1	ELBO Safe	RA Safe	Rank Δ
VCIP	0.973	44.4%	39.4%	84.6%	95.6%	+2.0
ACTIN	0.676	5.2%	4.4%	52.2%	75.8%	−12.9
CT	0.544	3.2%	2.6%	47.8%	69.8%	−15.0
CRN	0.690	6.0%	5.2%	53.4%	76.4%	−12.9
RMSN	0.578	2.8%	2.0%	49.0%	72.2%	−13.2

6.5 Hard Filter vs. Soft Constraint

We compare hard filtering against two soft-constraint alternatives: (1) *fixed-λ Lagrangian*: $\bar{a}^* = \arg\min_{\bar{a}} \mathrm{ELBO}(\bar{a}) + \lambda(1 - J_{\mathrm{RA}}(\bar{a}))$ for $\lambda \in \{0.1, \ldots, 50\}$—the direct analogue of constrained policy optimization [Achiam et al., 2017, Tessler et al., 2019]; and (2) *PID-Lagrangian* (per-patient adaptive λ): binary search for the minimum λ that selects a feasible candidate—the analogue of responsive safety methods [Stooke et al., 2020] in our offline setting.

Hard filtering Pareto-dominates fixed-λ: at comparable safety ($\sim$86%), it achieves 32.0% Top-1 vs. 25.9% for the best fixed λ. The PID-Lagrangian is *equivalent* to hard filtering: at every horizon, both achieve identical safety and Top-1 (e.g., 91.6% safety, 46.4% Top-1 at $\tau = 8$). This equivalence is not coincidental—in an offline selection setting with binary feasibility, the minimum λ that redirects selection to a feasible candidate exactly implements the hard filter. The mechanism for fixed-λ's inferiority is that a global penalty degrades ELBO ranking quality among already-safe sequences, while hard filtering preserves the original ranking within $\mathcal{F}_{\mathrm{RA}}$ (confirming Proposition 5).

6.6 Ablations

We summarize four key ablations; full results (target set sensitivity, κ hardness, reach-vs-avoid decomposition, k-expansion) are in Appendix G.

Midpoint baseline. A natural alternative is to set $Y^* = \mathrm{midpoint}(\mathcal{T})$ and use standard ELBO selection without RA filtering. On Cancer ($\gamma = 4$, oracle trajectories), the midpoint oracle achieves higher in-target rates (96–99%) than RA (94–97%) but at substantially worse Top-1 accuracy (18–58% vs. 17–60%) because midpoint distance and ground-truth loss are different objectives. Critically, the midpoint baseline *does not enforce intermediate safety*—its safety rates equal unconstrained ELBO (Table 19, Appendix M). RA-constrained selection is the only method achieving both high in-target *and* high intermediate safety.

Target set size. The method offers a smooth trade-off (Table 8, Appendix G): narrow targets ($\mathcal{T}_u = 1.5$) yield +27 pp safety at $\tau = 8$ but only 1.7% feasibility; broad targets ($\mathcal{T}_u = 10$) become near-no-ops. Our default ($\mathcal{T}_u = 3.0$) balances safety gain (+9–23 pp) with feasibility (10–54%).

$\varepsilon_{\mathrm{VI}}$ estimation. We estimate $\hat{\varepsilon}_{\mathrm{VI}} = 0.23$ (γ=1) to 0.11 (γ=4) using rank MAE (Appendix B). The bound is non-vacuous for feasibility-differing pairs and tightest where RA filtering is most valuable.

Oracle vs. model-predicted trajectories. On Cancer, model-predicted RA filtering yields near-zero feasibility ($< 0.5\%$) because the decoder produces near-constant predictions regardless of treatment sequence—a consequence of the latent space being action-invariant (KL divergence between latent distributions under different treatments $\approx 10^{-5}$; Section D). We tested three decoder modifications—heteroscedastic (NLL) loss, wider architecture ($[64, 32]$ hidden units), and MC-dropout—and none

recovered discrimination (Table 22, Appendix P). This establishes that the bottleneck is *upstream* of the decoder: the variational encoder does not encode action-dependent information into the latent state.

On MIMIC-III, model-based predictions straddle the DBP constraint boundary, giving genuine discriminating power (22–33% feasibility). We propose **model feasibility rate** as a deployment readiness diagnostic: when near zero, the model's trajectory predictions do not span constraint boundaries and oracle trajectories are needed; when non-trivial (>10%), model-predicted filtering provides meaningful discrimination.

These findings define a *three-regime framework* for safety in counterfactual planning: (1) **Oracle safety**: uses ground-truth trajectories (available only in simulation); provides an upper bound on achievable safety. (2) **Optimistic safety**: uses model-predicted trajectories; effective when predictions span constraint boundaries (MIMIC) but not when compressed (Cancer). (3) **Certified safety**: provides distribution-free guarantees via conformal prediction; requires well-calibrated trajectory models (Appendix N). The model feasibility rate determines which regime applies in a given deployment context.

7 Discussion and Conclusion

When does RA help most? The benefit scales with confounding strength (Table 2): at $\gamma = 1$, ELBO already selects safe plans; at $\gamma \geq 3$, RA filtering provides 12–15 pp average safety improvement across horizons—most valuable where planning is most challenging.

Failure modes and cost of safety. Unsafe ELBO selections are dominated by conservative undershoot at short τ and toxic paths at long τ (Appendix I); most patients (54–65/100) are "sometimes unsafe," confirming broad need for filtering. The Top-1 cost (1–15 pp) is distributed across 82/100 patients (Appendix H)—a fundamental property of hard constraints. RA scoring alone fails as a ranker (Top-1 $\approx 1\%$); ELBO ranking and RA feasibility are complementary.

Toward clinical guardrails. Post-hoc safety filtering is analogous to shielding in safe RL [Alshiekh et al., 2018] and rejection sampling in RLHF [Bai et al., 2022]. Safety thresholds can be adjusted by clinicians without retraining, and the same model can serve different constraint profiles.

Limitations and future work. (1) $\mathcal{T}$ and $\mathcal{S}$ require domain expertise (Table 8 shows robustness to threshold choice). (2) On MIMIC-III, counterfactuals are unobservable; matched-treatment analysis shows a modest +2–3 pp observed in-target improvement. A blinded evaluation by two ICU intensivists confirms RA plans are preferred (combined $d = 0.69$, 83% preference; Appendix O.6). (3) Feasibility drops at long horizons; expanding k mitigates this (feasibility scales linearly). (4) Sequential ignorability is assumed; Theorem 6 provides formal sensitivity bounds, tightened to $\Gamma_{\max} > 30$ under time-invariant confounders (Proposition 7). (5) On Cancer, model-predicted trajectories yield near-zero feasibility due to action-invariant latent representations (Section 6.6); three decoder modifications fail to recover discrimination, identifying the latent encoder as the bottleneck. On MIMIC, model-based filtering is effective. We propose model feasibility rate as a deployment readiness diagnostic. (6) The MIMIC evaluation uses DBP as a single outcome; clinician feedback suggests multi-outcome targets (e.g., MAP + SpO_2) would improve clinical relevance (Appendix O.6). Future work includes multi-outcome targets, action-aware latent encoders, and adaptive threshold selection.

Reproducibility. Code for reach-avoid constrained selection, all experiments, and analysis scripts will be released upon acceptance. The Cancer simulator [Bica et al., 2020c] is publicly available. MIMIC-III [Johnson et al., 2016] is available via PhysioNet credentialed access. All experiments use 5 random seeds; hyperparameters, thresholds, and hardware specifications are detailed in Appendix G.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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   Justification: Section 7 ("Limitations and future work") discusses five limitations: domain expertise for threshold selection, unobservable counterfactuals on MIMIC-III, feasibility drop at long horizons, the sequential ignorability assumption (with formal sensitivity analysis in Theorem 6), and oracle trajectory dependence on the Cancer simulator.

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   Justification: Theorem 1 states two assumptions (Assumptions 1 and 2), with proof sketch in Section 5 and full proof in Appendix A. Theorem 6 adds a third assumption (Assumption 3), with proof sketch in Section 5.3 and full proof in Appendix J. Propositions 3–4 are proved in Appendix L.

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A Full Proof of Theorem 1

Proof. **Part (a): RA ranking preservation.**

Define the true RA probability under p_θ:

$$P(\bar{a}) = \mathbb{E}_{p_\theta}\left[\mathbf{1}\{Y_{t+\tau} \in \mathcal{T}\} \prod_{s=t}^{t+\tau} \mathbf{1}\{Y_s \in \mathcal{S}\} \;\middle|\; \bar{H}_t, \bar{a}\right].$$

Let $P_q(\bar{a})$ denote the same quantity under the variational distribution q_ϕ. Since $f(\mathbf{Y}) := \mathbf{1}\{Y_{t+\tau} \in \mathcal{T}\} \prod_s \mathbf{1}\{Y_s \in \mathcal{S}\} \in [0,1]$, the variational characterization of total variation gives:

$$|P(\bar{a}) - P_q(\bar{a})| = |\mathbb{E}_{p_\theta}[f] - \mathbb{E}_{q_\phi}[f]| \leq \mathrm{TV}(p_\theta, q_\phi) \cdot \|f\|_\infty \leq \varepsilon_{\mathrm{VI}}.$$

Next, the soft-indicator approximation. Let $\hat{P}(\bar{a}) = \mathbb{E}_{q_\phi}[g_{\mathcal{T}}(Y_{t+\tau}) \prod_s g_{\mathcal{S}}(Y_s)]$. Since $|g_{\mathcal{T}}(y) - \mathbf{1}_{\mathcal{T}}(y)| \leq \varepsilon_{\text{soft}}$ and $|g_{\mathcal{S}}(y) - \mathbf{1}_{\mathcal{S}}(y)| \leq \varepsilon_{\text{soft}}$ pointwise, and for products of n terms bounded in $[0, 1]$ with pointwise error $\leq \varepsilon$:

$$\left| \prod_{i=1}^{n} g_i - \prod_{i=1}^{n} h_i \right| \leq \sum_{i=1}^{n} |g_i - h_i| \leq n \cdot \varepsilon,$$

(by telescoping and the bound $|\prod_{j>i} g_j| \leq 1$), we get $|P_q(\bar{a}) - \hat{P}(\bar{a})| \leq (\tau + 1)\varepsilon_{\text{soft}}$.

Combining by triangle inequality:

$$|\hat{P}(\bar{a}) - P(\bar{a})| \leq \varepsilon_{\text{VI}} + (\tau + 1)\varepsilon_{\text{soft}}.$$

For two sequences $\bar{a}_1, \bar{a}_2$:

$$|\hat{P}(\bar{a}_1) - \hat{P}(\bar{a}_2)| \geq |P(\bar{a}_1) - P(\bar{a}_2)| - |\hat{P}(\bar{a}_1) - P(\bar{a}_1)| - |\hat{P}(\bar{a}_2) - P(\bar{a}_2)| \tag{24}$$

$$\geq |P(\bar{a}_1) - P(\bar{a}_2)| - 2[\varepsilon_{\text{VI}} + (\tau + 1)\varepsilon_{\text{soft}}]. \tag{25}$$

If $|P(\bar{a}_1) - P(\bar{a}_2)| > 2\varepsilon_{\text{VI}} + 2(\tau + 1)\varepsilon_{\text{soft}}$, then $|\hat{P}(\bar{a}_1) - \hat{P}(\bar{a}_2)| > 0$ and the ranking sign is preserved. □

Part (b): Point-target ranking threshold.

For point-target ranking, define $d(\bar{a}) = \mathbb{E}_{p_\theta}[\|Y_{t+\tau}[\bar{a}] - Y^*\|^2]$. The function $f(Y) = \|Y - Y^*\|^2$ takes values in $[0, C_d]$ where $C_d = \sup_{\bar{a}} \|Y_{t+\tau}[\bar{a}] - Y^*\|^2_{L^\infty(p_\theta)}$ is the essential supremum over the model's outcome support. By the variational characterization of total variation applied to $f/C_d \in [0, 1]$:

$$|d(\bar{a}) - d_q(\bar{a})| = |\mathbb{E}_{p_\theta}[f] - \mathbb{E}_{q_\phi}[f]| \leq \text{TV}(p_\theta, q_\phi) \cdot C_d \leq \varepsilon_{\text{VI}} \cdot C_d.$$

The same pairwise argument as Part (a) gives ranking preservation when:

$$|d(\bar{a}_1) - d(\bar{a}_2)| > 2\varepsilon_{\text{VI}} \cdot C_d.$$

Since C_d scales with the squared diameter of the outcome support (e.g., $C_d = V_{\max}^2$ for cancer volume $\in [0, V_{\max}]$), this threshold can be much larger than the RA threshold of $2\varepsilon_{\text{VI}}$—precisely the clinical regime where outcomes have high spread. □

Part (c): Strict improvement via boundary mass decomposition.

Decompose the outcome space into three regions relative to $\mathcal{T}$: the *interior* $\mathcal{T}_\epsilon^\circ$ (distance $> \epsilon$ from $\partial\mathcal{T}$), the *exterior* $(\mathbb{R}^d \setminus \mathcal{T})_\epsilon$ (distance $> \epsilon$ from $\partial\mathcal{T}$), and the *boundary strip* $\partial_\epsilon\mathcal{T}$ (within ϵ of $\partial\mathcal{T}$). When the boundary mass $\beta = \sup_{\bar{a} p_\theta(Y_{t+\tau}[\bar{a}] \in \partial_\epsilon \mathcal{T}) \ll 1}$, most outcomes are unambiguously in or out of $\mathcal{T}$.

For pairs where $\bar{a}_1$ is feasible and $\bar{a}_2$ is not, and both have low boundary mass:

- RA margin: $m_{\text{RA}} = |P(\bar{a}_1) - P(\bar{a}_2)| \geq (1 - \beta) - \beta = 1 - 2\beta \approx 1$.
- ELBO margin: the sequences may have nearly identical expected distances $d(\bar{a}_1) \approx d(\bar{a}_2)$ if their outcomes are on opposite sides of $\partial\mathcal{T}$ at similar distances from Y^*, giving $m_{\text{ELBO}} \approx 0$.

Thus $m_{\text{RA}} \gg m_{\text{ELBO}}/C_d$ when boundary mass is small. □ □

B Justification of ε_{VI} Proxy

In Section 6.6 we estimate ε_{VI} using rank mean absolute error (rank MAE) between ELBO-predicted and ground-truth loss rankings. Here we formalize the connection between rank MAE and total variation distance, showing that our proxy yields a conservative (upper) bound on the ranking-relevant component of ε_{VI}.

Setup. For a patient with k candidate sequences $\bar{a}_1, \ldots, \bar{a}_k$, let σ^* denote the ground-truth ranking (by true loss) and $\hat{\sigma}$ the ELBO ranking (by model loss). The rank MAE is:

$$\text{rankMAE} = \frac{1}{k} \sum_{i=1}^{k} |\hat{\sigma}(i) - \sigma^*(i)|.$$

Connection to ranking displacement. The rank MAE is the ℓ_1 displacement between permutations, known as *Spearman's footrule* D [Diaconis and Graham, 1977]. By the Diaconis–Graham inequality, the footrule satisfies:

$$K_\tau \;\leq\; D \;\leq\; 2K_\tau, \tag{26}$$

where K_τ is the number of pairwise inversions (Kendall tau distance). Thus rank MAE and pairwise inversion rate are equivalent up to a factor of 2.

From inversions to TV. For any pair $(\bar{a}_i, \bar{a}_j)$, Theorem 1(a) shows the pairwise ELBO ranking is preserved when the true probability margin exceeds $2\varepsilon_{\text{VI}}$. Contrapositively, an inversion implies

$$|P(\bar{a}_i) - P(\bar{a}_j)| \leq 2\varepsilon_{\text{VI}} + 2(\tau + 1)\varepsilon_{\text{soft}}(\kappa).$$

For hard filtering ($\kappa \geq 5$, $\varepsilon_{\text{soft}} \approx 0$), the inversion rate across all candidate pairs provides a lower bound on the fraction of pairs whose probability margins are within $2\varepsilon_{\text{VI}}$ of zero. The rank MAE thus captures the *practical consequence* of the variational gap: how much the ranking is displaced.

Conservative bound. We use rank MAE directly as $\hat{\varepsilon}_{\text{VI}}$ because:

- Rank MAE $\in [0, 1]$ after normalization by $(k-1)/2$, matching the range of TV distance.
- Rank MAE captures all sources of ranking error (variational gap, finite candidate pool, model misspecification), not just TV. It is therefore an *upper bound* on the TV-induced ranking displacement.
- The bound is non-vacuous: at $\gamma = 4$, $\hat{\varepsilon}_{\text{VI}} = 0.114$, so the margin condition $|P(\bar{a}_1) - P(\bar{a}_2)| > 2 \times 0.114 = 0.228$ is easily satisfied for feasibility-differing pairs (where $|P| = 1$).

Limitations. This proxy conflates model error (which TV captures) with other ranking distortions (e.g., finite-sample effects, non-identifiability). A tighter bound would require direct estimation of $\text{TV}(p_\theta, q_\phi)$ via density ratio estimation [Sugiyama et al., 2012], which is intractable for the high-dimensional trajectory distributions in our setting. Our rank MAE proxy is a *practical diagnostic* rather than a formal statistical bound; we present it transparently as such.

C Extended Related Work

We provide a comprehensive survey of the literatures that intersect with our contribution: counterfactual outcome estimation, variational methods for causal inference, dynamic treatment regimes, safe sequential decision-making, and reach-avoid control. Below we discuss each in detail and clarify the positioning of our work.

C.1 Counterfactual Outcome Estimation Over Time

Estimating individualized treatment effects from observational data is a rapidly growing area at the intersection of causal inference and deep learning. Foundational work by Shalit et al. [2017] established generalization bounds for individual treatment effect estimation via balanced representations, showing that controlling distributional distance between treated and control groups is key to reducing estimation error. This line was extended to generative modeling: CEVAE [Louizos et al., 2017] used variational autoencoders to simultaneously infer hidden confounders and causal effects, while GANITE [Yoon et al., 2018] employed generative adversarial networks to capture uncertainty in counterfactual distributions. Alaa and van der Schaar [2017] took a Bayesian nonparametric approach with multi-task Gaussian processes for individualized treatment effects.

The *longitudinal* setting—where treatments are administered sequentially and outcomes evolve over time—poses additional challenges from time-varying confounding. Marginal structural models [Robins et al., 2000] use inverse probability weighting to handle this; CRN [Bica et al., 2020a] learns balanced representations adversarially; RMSN [Lim, 2018] combines RNNs with stabilized weights; and G-Net [Li et al., 2021] implements g-computation with recurrent networks. The Causal Transformer [Melnychuk et al., 2022] adapts attention-based architectures for this setting, while ACTIN [Seedat et al., 2022] models continuous-time treatment effects via neural controlled differential equations. SyncTwin [Qian et al., 2021] constructs synthetic control twins from longitudinal pre-treatment data to estimate effects at the individual level.

Addressing hidden confounding in sequential settings, the Time Series Deconfounder [Bica et al., 2020b] leverages factor models over treatment assignments to infer substitutes for unobserved confounders, while Frauen et al. [2023] derive sharp sensitivity bounds applicable to time-varying treatments under generalized marginal sensitivity models. Recent work continues to advance this frontier: Causal CPC [Kim et al., 2024] incorporates contrastive predictive coding into counterfactual regression, and Wu et al. [2024] propose conditional generative models (diffusion, CVAE) for high-dimensional counterfactual outcomes under time-varying treatments.

All of these methods estimate outcomes for *given* treatment sequences but do not optimize over them. VCIP [Melnychuk et al., 2025] bridges this gap by selecting intervention sequences via ELBO minimization toward a point target. Our work keeps the VCIP estimation infrastructure intact and adds a safety-aware selection layer—the first to incorporate *path constraints* into variational counterfactual planning.

C.2 Variational and Generative Models for Causal Inference

Variational autoencoders [Kingma and Welling, 2014] have become a natural backbone for causal inference under latent confounding. CEVAE [Louizos et al., 2017] pioneered this by combining a VAE with the causal graph structure, using proxy variables to infer hidden confounders. Deep Structural Causal Models [Pawlowski et al., 2020] extended this to full Pearl-style counterfactual reasoning by learning invertible mechanisms with normalizing flows, enabling all three levels of the causal hierarchy (association, intervention, counterfactual). Zhang et al. [2021] explicitly disentangled instrumental, confounding, and adjustment latent factors for treatment effect estimation.

Deep end-to-end causal inference (DECI; Geffner et al. 2022) jointly learns causal structure and parameters with a variational framework. Variational Causal Inference [VCI; Wu and Bareinboim, 2025] derives an ELBO for individual-level counterfactual likelihoods in a static setting and introduces latent disentanglement via $D_{\mathrm{KL}}[q(z|y,t,x)\|q(z|y',t',x)]$. We draw on VCI's insight that latent action-sensitivity is a diagnostic for counterfactual model quality (Appendix E), adapting it to the sequential setting. Our diagnostic reveals that VCIP's latent space is nearly action-invariant ($D_{\mathrm{KL}} \approx 10^{-5}$), with action-outcome coupling residing entirely in the decoder—a structural property that explains why RA filtering operates effectively as a post-hoc method without model modification.

C.3 Dynamic Treatment Regimes

The DTR literature [Murphy, 2003, Chakraborty and Moodie, 2014, Hernán and Robins, 2020] formalizes sequential treatment decisions under the potential outcomes framework. Classical methods include Q-learning, A-learning, and outcome-weighted learning for estimating optimal decision rules mapping patient states to treatment actions. Nie et al. [2021] address the "when-to-treat" problem with advantage doubly robust estimators and welfare regret bounds under sequential ignorability, while batch policy learning methods [Le et al., 2019] handle constraints in offline settings with finite samples.

Most DTR methods learn *decision rules* (state $\rightarrow$ action mappings) that optimize expected outcomes over the full trajectory. In contrast, VCIP and our extension operate in a *counterfactual planning* mode: given a specific patient history, they evaluate and rank complete intervention sequences. The reach-avoid formulation adds the constraint that trajectories must remain safe—a concern shared with constrained DTR work but addressed here through a distinct mechanism. Where constrained DTR methods embed safety into the policy learning objective (requiring retraining when constraints change), our post-hoc filtering approach is decoupled from the base model: constraints can be modified at deployment time without any retraining.

C.4 Safe and Constrained Sequential Decision-Making

Safety in sequential decision-making has been approached from multiple angles. The constrained MDP framework [Altman, 1999] provides the theoretical foundation, where the agent maximizes reward subject to expected cost constraints. Constrained Policy Optimization [CPO; Achiam et al., 2017] introduced the first general-purpose policy search algorithm for constrained RL with near-constraint satisfaction guarantees at each iteration, using trust region methods. Reward Constrained Policy Optimization [RCPO; Tessler et al., 2019] takes a Lagrangian approach with multi-timescale updates, while CUP [Yang et al., 2022] provides tighter constraint satisfaction via constrained update projections.

Lyapunov-based methods [Chow et al., 2018] provide safety guarantees through stability certificates, and safety layers [Dalal et al., 2018] project unsafe actions onto a safe set in real time. The comprehensive survey by García and Fernández [2015] categorizes safe RL approaches into those modifying the optimization criterion, the exploration process, or both.

The concept of *shielding* [Alshiekh et al., 2018]—a post-hoc safety monitor that overrides unsafe actions proposed by an RL agent—is the closest conceptual analogue to our approach. Where a shield intercepts individual actions, our RA filter intercepts entire candidate treatment sequences, rejecting those that violate reach-avoid feasibility. A particularly relevant recent direction is *offline* safe RL, which learns safe policies from fixed datasets without environment interaction—closely paralleling our offline planning setting. FISOR [Zheng et al., 2024] uses Hamilton-Jacobi reachability within a diffusion model for safe offline policy learning with hard constraints, achieving zero-violation performance. This is conceptually related to our approach: both use reachability concepts for safety, but FISOR learns a policy while we filter pre-computed candidate sequences.

Chance-constrained optimization [Nemirovski and Shapiro, 2006, Calafiore and Campi, 2006] provides a complementary perspective: constraints of the form $\Pr(g(x, \xi) \leq 0) \geq 1 - \alpha$ require that safety holds with high probability over uncertain parameters ξ. Our feasibility check (Eq. 1) can be viewed as an empirical chance constraint, where feasibility over k sampled trajectories approximates a population-level safety guarantee. Robust MDPs [Iyengar, 2005, Nilim and El Ghaoui, 2005] address model uncertainty by optimizing worst-case performance over an ambiguity set of transition models—a setting related to our Theorem 1, which bounds the effect of distributional mismatch (variational gap) on ranking.

Our setting differs fundamentally from online safe RL in that we have no opportunity for environment interaction or policy correction. We select among candidate plans evaluated on a learned model, making a *filter-based* approach natural rather than constrained policy optimization. This is both a limitation (we cannot explore beyond the candidate set) and a strength (no risk of unsafe exploration, no retraining required, and the filter is model-agnostic).

C.5 Reach-Avoid Analysis and Control

The reach-avoid problem—steering a dynamical system into a target set while avoiding unsafe regions—has deep roots in control theory. The Hamilton-Jacobi (HJ) formulation [Mitchell et al., 2005] computes backward reachable sets by solving a PDE, providing formal safety guarantees but suffering from exponential computational cost in the state dimension. Fisac et al. [2015] extended this to time-varying dynamics, targets, and constraints. Altschuler and Parrilo [2023] offered an alternative via sum-of-squares optimization for polynomial systems.

DeepReach [Bansal and Tomlin, 2021] bridges the gap between classical HJ reachability and modern deep learning, using sinusoidal neural networks to approximate value functions for high-dimensional reachability problems, with computational cost scaling with the complexity of the reachable tube rather than grid resolution. ISAACS [Hsu et al., 2023] combines reach-avoid games with actor-critic methods for adversarial safety.

We import the reach-avoid concept into the variational counterfactual planning framework, but with a key simplification: rather than solving a continuous reachability PDE, we perform a *discrete feasibility check* over a finite set of candidate treatment sequences. Each sequence's trajectory is evaluated against the target and safety sets using soft indicator functions, converting the reach-avoid problem from continuous optimal control to combinatorial filtering. This discrete formulation is computationally trivial (linear in the number of candidates and horizon length) and avoids the curse

of dimensionality inherent in HJ methods, at the cost of being limited to the candidate set generated by the base planner.

C.6 Safety in Clinical AI and Treatment Optimization

AI-driven treatment optimization in clinical settings faces unique safety challenges. The AI Clinician [Komorowski et al., 2018] pioneered the use of RL for sepsis treatment in the ICU, discretizing fluid and vasopressor dosing into 25 actions and showing that policies closer to the AI's recommendations correlated with improved 90-day survival. Raghu et al. [2017] explored continuous state-space RL for the same problem. However, these approaches do not explicitly enforce safety constraints during planning—a limitation that has motivated subsequent work on constrained clinical RL.

In oncology, treatment planning must balance tumor control against toxicity. The cancer simulator [Geng et al., 2017, Bica et al., 2020c] used in our experiments models this tradeoff: chemotherapy reduces tumor volume but accumulates toxicity. Our reach-avoid formulation directly operationalizes clinical safety constraints (e.g., tumor volume must stay below a dangerous threshold at every intermediate step, chemotherapy dosage must not exceed toxicity limits) that are implicit in clinical practice but absent from ELBO-based planning objectives.

The post-hoc filtering paradigm we propose is related to the broader concept of *guardrails* in AI safety: rather than training the model to be safe, we add an external safety check on its outputs. This separation of concerns—quality from the base model, safety from the filter—has practical advantages in clinical deployment: safety thresholds can be adjusted by clinicians without requiring model retraining, and the same base model can operate under different constraint profiles for different patient populations or clinical contexts.

D RA-Aware Retraining: Null Result

We tested whether retraining VCIP with RA-motivated losses improves constrained selection. We trained a `ReachAvoidVAEModel` with: (1) weighted intermediate loss ($\lambda_{\text{int}} = 0.5$, $\lambda_{\text{term}} = 1.0$) encouraging accurate per-step predictions, and (2) VCI-inspired disentanglement ($\lambda_{\text{disent}} = 0.1$) encouraging the latent space to separate by action. Training: 5 seeds $\times$ 2 gammas = 10 runs on Vast.ai (2$\times$ RTX 3090, $\sim$1.5h, $\sim$\$3).

Table 5: Vanilla VCIP vs. RA-retrained VCIP ($\gamma = 4$). All metrics are identical within noise ($\pm$0.002 GRP, $\pm$1.4 pp Top-1).

τ	Van. GRP	RA GRP	Van. Cstr-Top1	RA Cstr-Top1	Van. Safe	RA Safe
2	0.922	0.924	17.4%	18.0%	94.4%	94.4%
4	0.963	0.963	39.4%	38.6%	95.6%	95.6%
6	0.981	0.982	54.2%	55.0%	96.2%	96.2%
8	0.984	0.984	60.4%	60.4%	97.0%	97.0%

All metrics are identical within noise (Table 5). The reason: on Cancer, RA filtering uses *ground-truth simulator trajectories* for feasibility assessment. The model only determines the ELBO ranking within the feasible set, which is already excellent (GRP > 0.92). Retraining cannot improve what the model does not control. This confirms that RA-constrained selection is a *pure post-hoc method* requiring no retraining—a strength, not a limitation.

E VCI Consistency Diagnostic

Adapting VCI's latent disentanglement insight [Wu and Bareinboim, 2025] to the sequential setting, we measured $D_{\text{KL}}[q(Z_s \mid \bar{H}_t, \bar{a}_{\text{obs}}) \| q(Z_s \mid \bar{H}_t, \bar{a}_{\text{alt}})]$ for 100 individuals, 20 alternative action sequences, across all τ and 5 seeds ($\gamma = 4$). The result: $D_{\text{KL}} \approx 10^{-5}$ everywhere.

VCIP's latent space is nearly *action-invariant*: swapping the entire action sequence barely changes the posterior $q(Z_s)$. Action-outcome coupling lives entirely in the *decoder* ($p_\theta(Y_s|Z_s, a_s)$), not in the latent representations. This architectural property explains why RA filtering works post-hoc without

Table 6: VCI consistency diagnostic: mean D_{KL} between observed and alternative action posteriors ($\gamma = 4$, 5 seeds).

	$\tau = 2$	$\tau = 4$	$\tau = 6$	$\tau = 8$
Mean D_{KL}	1.2×10^{-5}	3.0×10^{-5}	4.0×10^{-5}	4.5×10^{-5}

any latent-space modifications: ELBO differences between sequences come from the decoder's output, which does depend on actions, even though the latent space does not.

F Additional Experimental Details

VCIP model configuration. All Cancer experiments use the default VCIP hyperparameters from Melnychuk et al. [2025]: latent dimension $d_z = 16$, hidden dimension $d_h = 16$, 100 training epochs, batch size 1024 ($\gamma = 4$) or 256 ($\gamma = 1$). MIMIC experiments use the same architecture with dataset-specific adjustments (outcome dimension, treatment space).

Candidate sequence generation. Following VCIP, we generate $k = 100$ candidate sequences per patient by perturbing the observed treatment sequence. Perturbation involves flipping individual treatment decisions with probability proportional to the inverse of the behavioral policy, ensuring diversity while maintaining clinical plausibility.

Compute budget. Cancer model training: $\sim$52 hours on 4$\times$ NVIDIA RTX 2060 ($\sim$\$10). MIMIC model training: $\sim$19.5 hours on 2$\times$ NVIDIA RTX 3090 ($\sim$\$8). Phase 2 RA-retraining: $\sim$1.5 hours on 2$\times$ RTX 3090 ($\sim$\$3). RC6/E4 GPU evaluation: $\sim$1 hour on 2$\times$ RTX 3090. All RA-constrained selection analyses (Phase 1B, ablations) are purely offline and require negligible compute ($<$ 1 minute per configuration on a single CPU).

Standard deviations. Table 7 reports standard deviations across seeds for key metrics at $\gamma = 4$.

Table 7: Standard deviations across 5 seeds ($\gamma = 4$).

τ	GRP std	ELBO Top-1 std	RA Top-1 std	RA Safety std
2	0.008	2.1%	2.3%	1.8%
4	0.005	3.4%	3.8%	1.2%
6	0.002	2.8%	3.1%	0.9%
8	0.002	2.2%	2.6%	0.7%

G Ablation Tables

Table 8: Target set size sensitivity ($\gamma = 4$). Thresholds: $\mathcal{T}_{\text{upper}}$ (target volume), $\mathcal{S}_{\text{vol}}$ (safety volume), $\mathcal{S}_{\text{chemo}}$ (safety dosage).

Config	$\mathcal{T}_u$	$\mathcal{S}_v$	$\mathcal{S}_c$	τ=2 Feas	τ=2 SafeΔ	τ=8 Feas	τ=8 SafeΔ	τ=8 Top1Δ
Narrow	1.5	6.0	3.0	26.0%	+15.8	1.7%	+27.4	−20.0
Moderate	3.0	12.0	5.0	53.6%	+9.4	10.3%	+22.6	−15.4
Broad	6.0	20.0	8.0	95.9%	+4.6	74.6%	+3.2	−1.6
V. Broad	10.0	30.0	12.0	98.6%	+1.4	99.6%	+1.4	−0.2

Table 8 reveals a smooth, continuously tunable trade-off between safety and selection quality. Narrow targets ($\mathcal{T}_u = 1.5$) eliminate nearly all candidates at long horizons (1.7% feasibility at $\tau = 8$), yielding the largest safety improvement (+27.4 pp) but at a substantial Top-1 cost (−20.0 pp). At the other extreme, very broad targets ($\mathcal{T}_u = 10.0$) become near no-ops: 99.6% feasibility with only +1.4 pp safety gain. Our moderate default ($\mathcal{T}_u = 3.0$) occupies the favorable region of this frontier, providing substantial safety improvement (+22.6 pp at $\tau = 8$) while retaining meaningful feasibility (10.3%). The monotonic relationship between threshold strictness and both safety gain and Top-1

cost confirms that the method is not brittle to threshold choices—practitioners can select an operating point appropriate to their clinical risk tolerance.

Table 9: Sigmoid hardness κ sensitivity ($\gamma = 4$, averaged across τ).

κ	1	2	5	10	50	100
Agreement w/ hard filter	96.8%	99.5%	99.9%	100%	100%	100%
Top-1	42.3%	42.9%	42.9%	42.9%	42.9%	42.9%
Safety	93.7%	95.3%	95.8%	95.8%	95.8%	95.8%

Table 9 validates the soft indicator approximation (Assumption 2). For $\kappa \geq 5$, the soft sigmoid agrees with the hard binary filter at $> 99.9\%$, and all downstream metrics (Top-1 accuracy and safety rate) are identical to the hard-filter case. Even at $\kappa = 1$, agreement is already 96.8% with only marginal degradation in safety (93.7% vs. 95.8%). This confirms that $\varepsilon_{\text{soft}}(\kappa) \approx 0$ for all practical purposes, meaning the theoretical guarantees from Theorem 1 hold without requiring extreme hardness values.

Table 10: Reach-only vs. reach-avoid decomposition ($\gamma = 4$).

Method	τ=2 Top-1	τ=2 In-T	τ=2 Safe	τ=8 Top-1	τ=8 In-T	τ=8 Safe
Unconstrained	18.8%	85.0%	85.0%	75.8%	85.6%	85.6%
Reach-only	17.2%	96.2%	96.0%	66.4%	99.2%	98.4%
Avoid-only	18.8%	85.0%	85.0%	68.2%	87.0%	87.0%
Reach-avoid	17.4%	94.4%	94.4%	60.4%	97.0%	97.0%

Table 10 isolates the individual contribution of each constraint component. The reach (target) constraint is the primary driver of improvement: it lifts in-target rates from 85.0% to 96.2% at $\tau = 2$ and from 85.6% to 99.2% at $\tau = 8$, accounting for approximately 85% of the total gain. The avoid (safety) constraint alone is nearly a no-op at short horizons ($\tau = 2$: identical to unconstrained), which is expected since few intermediate states can violate safety over only two steps. At longer horizons ($\tau = 8$), avoid provides incremental value (87.0% vs. 85.6% safety) by catching trajectories that reach the target but pass through unsafe intermediate states. The Top-1 cost is roughly additive: reach-only costs $\sim$9 pp at $\tau = 8$ (75.8% $\rightarrow$ 66.4%), while the full reach-avoid costs $\sim$15 pp (75.8% $\rightarrow$ 60.4%). Both components are justified: reach provides the primary clinical benefit (keeping terminal outcomes in the target range), while avoid provides the formal path-safety guarantee that would be required in any real clinical deployment.

Table 11: ε_{VI} estimates and pairwise ranking preservation by γ.

γ	Rank MAE ($\hat{\varepsilon}_{\text{VI}}$)	Spearman ρ	ELBO preserved	RA preserved
1	0.225	0.418	60.8–70.6%	59.4–70.3%
2	0.155	0.687	71.3–81.9%	68.1–81.1%
3	0.128	0.804	77.5–84.4%	73.4–83.3%
4	0.114	0.847	79.0–86.9%	74.3–85.6%

Table 11 provides empirical estimates of the variational gap ε_{VI} using rank mean absolute error as a proxy. The gap *decreases* with confounding strength ($\hat{\varepsilon}_{\text{VI}} = 0.225$ at $\gamma = 1$ to 0.114 at $\gamma = 4$), indicating that VCIP's ELBO is better calibrated when treatment effects are stronger. This is a favorable property: the Theorem 1 bound is tightest precisely where RA filtering is most valuable. ELBO pairwise ranking preservation ranges from 61–87%, and the bound is non-vacuous: for pairs differing in feasibility status (margin $m = 1$), the condition $m > 2\varepsilon_{\text{VI}} \approx 0.23$–$0.45$ is easily satisfied. RA-preserved rates closely track ELBO-preserved rates, confirming that the feasibility filter does not introduce additional ranking distortion beyond what the variational approximation already entails.

Table 12 confirms that feasibility scales linearly with pool size: $|\mathcal{F}_{\text{RA}}| \approx k \cdot p$ where $p \approx 0.70$ is the per-candidate feasibility rate. Safety and in-target rates are stable across k. Top-1 accuracy decreases slightly (32.8% $\rightarrow$ 26.0%) because the oracle-best plan is harder to identify via ELBO ranking among a larger feasible set—a noise-in-ranking effect, not a method deficiency. Per-patient heterogeneity is large: some patients have near-zero feasibility while others exceed 90%, indicating that feasibility is driven by patient-specific disease dynamics rather than threshold choice.

Table 12: Candidate pool expansion (k) at $\gamma = 4$, $\tau = 8$ (moderate thresholds: $\mathcal{T}_u$=0.6, $\mathcal{S}_v$=5.0, $\mathcal{S}_c$=8.5). Feasibility rate is stable; the number of feasible candidates scales linearly with k.

k	Feasibility	$\lvert\mathcal{F}_{\mathrm{RA}}\rvert$	Cstr Top-1	In-Target	Safe
100	70.0%	69.8	32.8%	90.2%	90.2%
250	70.0%	174.9	27.4%	90.6%	90.6%
500	70.0%	350.0	26.2%	90.6%	90.6%
1000	71.0%	710.3	26.0%	90.6%	90.6%

H Heterogeneity Analysis

We analyzed whether the Top-1 accuracy cost of RA-constrained selection is concentrated in a few "hard" patients or broadly distributed ($\gamma = 4$, $\tau \in \{2, 4, 6, 8\}$).

Distribution of loss. Of 100 evaluation patients, 82 are "losers" (experience > 5 pp Top-1 decrease under RA constraint) and zero are "winners" (> 5 pp increase). The remaining 18 are approximately neutral. The top 20% of losers account for only 41% of total Top-1 loss—a near-uniform distribution rather than a concentrated one.

Loser characterization. Patients who lose the most Top-1 accuracy have lower average feasibility (68.2%) compared to neutral patients (higher feasibility). These are patients whose true optimal plan happens to violate safety constraints—a fundamental limitation of hard constraints, not a correctable estimation error.

Implication. Adaptive per-patient thresholds would not substantially reduce the Top-1 cost: the loss is not concentrated in identifiable subpopulations. The cost is the inherent price of the safety guarantee, distributed broadly across the patient population.

I Failure Taxonomy: When Do Counterfactual Planners Fail?

We systematically analyze when ELBO-optimal treatment selections are unsafe, complementing the summary in Section 7. All results use the paper's thresholds ($\mathcal{T}_u = 3.0$, $\mathcal{S}_v = 12.0$, $\mathcal{S}_c = 5.0$) on Cancer simulator data ($N = 100$ patients, 5 seeds, 4 gammas, 4 horizons).

I.1 Danger Rate

Table 13 reports the probability that the ELBO-optimal selection violates either the target or safety constraint. Two patterns emerge: (1) danger increases with horizon τ at all confounding levels, reaching 50.4% at γ=1, τ=8; (2) the *composition* of violations shifts with γ. At low confounding (γ=1), violations are overwhelmingly safety-only (path violations with acceptable terminal outcomes)—49.8% at τ=8. At high confounding (γ=4), target violations dominate at short horizons (13.6% at τ=2) while safety violations grow at longer horizons (11.2% at τ=8). This pattern confirms that confounding primarily degrades the ELBO's ability to select plans that reach the target, while path safety deteriorates independently as horizons lengthen.

I.2 Failure Mode Classification

We classify unsafe ELBO selections at γ=4 into three modes:

- **Mode A: Aggressive overshoot.** Terminal volume exceeds target *and* treatment intensity is above the median across all candidates. The model selects a high-treatment plan that still fails to control tumor growth.
- **Mode B: Conservative undershoot.** Terminal volume exceeds target with below-median treatment intensity. The model underestimates the required treatment dose.
- **Mode C: Toxic path.** Terminal volume is acceptable (≤ 3.0) but intermediate states violate safety. The plan reaches the target but through a dangerous trajectory.

Table 13: ELBO danger rate across γ and τ (500 patient-seed combinations per cell). "Target" = terminal cancer volume > 3.0; "Safety" = intermediate volume > 12.0 or chemo > 5.0.

γ	τ	Unsafe	Target only	Safety only	Both
1	2	0.8%	0.4%	0.4%	0.0%
1	4	9.0%	0.6%	8.2%	0.2%
1	6	29.2%	0.4%	28.6%	0.2%
1	8	50.4%	0.2%	49.8%	0.4%
4	2	15.0%	13.6%	0.0%	1.4%
4	4	16.8%	12.4%	1.4%	3.0%
4	6	20.2%	12.4%	5.0%	2.8%
4	8	25.6%	10.0%	11.2%	4.4%

Table 14 shows that Mode B (conservative undershoot) dominates at short horizons (100% at τ=2), while Mode C (toxic path) grows substantially at longer horizons (44% at τ=8). Mode A (aggressive overshoot) is rare ($\leq 13\%$), suggesting the ELBO's primary failure is *insufficient treatment* rather than excessive treatment. Unsafe selections have systematically higher mean terminal cancer volume (3.7–6.9) compared to safe selections (0.6–0.7), confirming that violations are not boundary cases but substantial deviations.

Table 14: Failure mode distribution at γ=4. Treatment intensity: mean total chemo dose for unsafe vs. safe selections.

τ	Mode A	Mode B	Mode C	Unsafe chemo	Safe chemo
2	0%	100%	0%	0.294	0.222
4	4%	88%	8%	0.435	0.340
6	13%	62%	25%	0.526	0.312
8	5%	52%	44%	0.422	0.326

I.3 Per-Patient Heterogeneity

At γ=4, aggregating across 5 seeds, the per-patient danger rate (fraction of seeds where the ELBO-optimal selection is unsafe) is broadly distributed:

- τ=2: 0 patients always unsafe, 54 sometimes unsafe, 44 always safe (mean rate: 0.15).
- τ=8: 0 patients always unsafe, 65 sometimes unsafe, 23 always safe (mean rate: 0.26).

No patient is *always* unsafe across all 5 seeds, but 54–65 out of 100 are "sometimes unsafe"—indicating that danger is broadly distributed across the patient population rather than concentrated in a few identifiable outliers. This supports the argument that safety filtering is broadly needed, not a correction for rare edge cases. The most dangerous patient at τ=8 (Patient 71) is unsafe in 4/5 seeds, with a mean terminal cancer volume of 2.02—just below the target threshold, indicating near-miss violations that are sensitive to random seed variation.

I.4 Cross-Model Failure Analysis

To assess whether failure modes are structural (inherent to the planning problem) or model-specific, we compare selection quality and failure patterns across all five baseline models at $\gamma = 4$.

Ranking quality varies dramatically across models. Table 15 reports the Spearman correlation between each model's ELBO rankings and the oracle true-loss rankings. VCIP substantially outperforms all other models: $\rho = 0.75$–0.90 across horizons, compared to CRN (0.12–0.33), RMSN (-0.07–0.29), ACTIN (0.15–0.28), and CT (≈ 0.00). VCIP's zero-regret rate (fraction of patients where the ELBO-best plan coincides with the oracle-best) ranges from 22% (τ=2) to 77% (τ=8), while CT achieves only 3–10%. This implies that RA-constrained selection is most valuable when paired with a model that provides meaningful surrogate rankings—without ranking quality, filtering selects randomly within the feasible set.

Table 15: Cross-model selection quality (Cancer, $\gamma = 4$). Spearman(ρ): rank correlation between model ELBO and oracle true loss. % zero: fraction of patients with zero selection regret. VCIP's ranking quality is 2–8$\times$ higher than all competitors.

	Spearman ρ (ELBO vs. oracle)				% zero regret			
Model	τ=2	τ=4	τ=6	τ=8	τ=2	τ=4	τ=6	τ=8
VCIP	0.75	0.84	0.89	0.90	22%	44%	64%	77%
CRN	0.33	0.25	0.17	0.12	4%	11%	14%	18%
CT	−0.01	−0.01	0.01	0.02	3%	7%	9%	10%
RMSN	0.29	0.12	−0.02	−0.07	10%	19%	19%	22%
ACTIN	0.28	0.25	0.19	0.15	4%	10%	14%	12%

Models learn nearly independent ranking functions. The pairwise Spearman correlation between model ELBO rankings (over $k = 100$ candidates, averaged across patients) is near zero for all model pairs: $\rho \in [-0.05, 0.22]$. VCIP–ACTIN show the highest agreement ($\rho \approx 0.19$), while CT's rankings are essentially independent of all other models. Despite learning from identical training data, these five architectures produce fundamentally different approximations of the treatment-outcome mapping, suggesting that ranking quality is highly sensitive to architectural choices.

Failure overlap is moderate, not universal. We define a patient as "dangerous" under a given model if, averaged across seeds, the ELBO-selected plan ranks in the worse half by true loss. No patient is dangerous under all 5 models simultaneously. At τ=2, 16 patients are dangerous under ≥ 3 models, and 10 are safe under all 5. The Spearman correlation of per-patient regret vectors across model pairs is moderate ($\rho = 0.1$–0.5), indicating partial structural overlap: the same patients tend to be *harder* across models, but each model fails in different ways. This supports the model-agnostic design of the RA filter: a post-hoc safety layer that does not depend on *which* model is used upstream.

J Sensitivity Analysis Under Hidden Confounding

This section provides the full proof of Theorem 6 and empirical sensitivity tables for the Cancer simulator.

J.1 Proof of Theorem 6

Proof. The proof proceeds in three steps.

Step 1: Per-step conditional TV bound. Fix a step s and history H_s. Under Assumption 3, the unobserved confounder U_s is associated with treatment a_s conditionally on H_s. The identified outcome distribution at step s integrates over U_s using its marginal:

$$P_0(Y_s \mid H_s, a_s) = \int P(Y_s \mid H_s, a_s, u)\, dP(u \mid H_s),$$

while the true distribution conditions on the confounder-treatment association:

$$P_\Gamma(Y_s \mid H_s, a_s) = \int P(Y_s \mid H_s, a_s, u)\, dP(u \mid H_s, a_s).$$

By Bayes' rule, $dP(u \mid H_s, a_s)/dP(u \mid H_s) = \pi_{\text{true}}(a_s \mid H_s, u)/\pi_{\text{obs}}(a_s \mid H_s) \in [1/\Gamma, \Gamma]$. The total variation between two distributions with Γ-bounded density ratio satisfies (see, e.g., Rosenbaum 2002):

$$\text{TV}(P_\Gamma(Y_s \mid H_s, a_s),\, P_0(Y_s \mid H_s, a_s)) \;\leq\; \varphi(\Gamma) := \frac{\Gamma - 1}{\Gamma + 1}. \tag{27}$$

This bound holds uniformly over all histories H_s and treatments a_s.

Step 2: Sequential maximal coupling. We construct a coupling of the τ-step trajectories $\bar{Y}^{(0)} \sim P_0$ and $\bar{Y}^{(\Gamma)} \sim P_\Gamma$. At each step $s = 1, \ldots, \tau$, conditional on the trajectories being coupled up to step $s-1$ (i.e., $Y^{(0)}_{1:s-1} = Y^{(\Gamma)}_{1:s-1}$), we apply a maximal coupling of $P_0(Y_s \mid H_s, a_s)$ and $P_\Gamma(Y_s \mid H_s, a_s)$. Under this coupling:

$$\Pr(Y^{(0)}_s \neq Y^{(\Gamma)}_s \mid \text{coupled at } s{-}1) = \text{TV}_s \leq \varphi(\Gamma).$$

The probability that the trajectories remain coupled through all τ steps is:

$$\Pr(\bar{Y}^{(0)} = \bar{Y}^{(\Gamma)}) \geq \prod_{s=1}^{\tau}(1 - \varphi(\Gamma)) = \left(\frac{2}{\Gamma + 1}\right)^{\tau}.$$

Therefore, the total variation between the joint trajectory distributions satisfies:

$$\mathrm{TV}(P_\Gamma(\bar{Y}),\, P_0(\bar{Y}))\ \leq\ 1 - \left(\frac{2}{\Gamma + 1}\right)^{\tau} =: \psi(\Gamma, \tau). \tag{28}$$

Step 3: Application to reach-avoid event. The reach-avoid event $E(\bar{a})$ is a measurable function of the full trajectory $\bar{Y}$, with indicator $\mathbf{1}\{E(\bar{a})\} \in [0, 1]$. By the variational characterization of total variation:

$$|P_\Gamma(E(\bar{a})) - P_0(E(\bar{a}))| = |\mathbb{E}_{P_\Gamma}[\mathbf{1}\{E\}] - \mathbb{E}_{P_0}[\mathbf{1}\{E\}]| \leq \mathrm{TV}(P_\Gamma, P_0) \leq \psi(\Gamma, \tau).$$

The population safety rate bound follows by linearity of expectation over patients: $\mathrm{SR}_\Gamma = \mathbb{E}_x[P_\Gamma(E(\bar{a}^*(x)))] \geq \mathbb{E}_x[P_0(E(\bar{a}^*(x)))] - \psi(\Gamma, \tau) = \mathrm{SR}_0 - \psi(\Gamma, \tau)$.

The breakdown point Eq. (22) is obtained by solving $\mathrm{SR}_0 - \psi(\Gamma, \tau) = 0$:

$$\left(\frac{2}{\Gamma + 1}\right)^{\tau} = 1 - \mathrm{SR}_0 \implies \Gamma_{\max} = \frac{2}{(1 - \mathrm{SR}_0)^{1/\tau}} - 1. \qquad \square$$

Remark on tightness. The sequential coupling bound $\psi(\Gamma, \tau) = 1 - (2/(\Gamma + 1))^\tau$ is strictly tighter than the union bound $\tau \cdot \varphi(\Gamma) = \tau(\Gamma - 1)/(\Gamma + 1)$ used in the proof sketch. For example, at $\Gamma = 1.5$, $\tau = 4$: the coupling bound gives $\psi = 0.59$ while the union bound gives 0.80. We present the tighter bound throughout.

J.2 Empirical Sensitivity Tables

Table 16 reports the guaranteed safety rate $\mathrm{SR}_\Gamma^{\mathrm{lower}} := \mathrm{SR}_0 - \psi(\Gamma, \tau)$ for the Cancer simulator at $\gamma = 4$ across planning horizons and confounding strengths.

Table 16: Sensitivity of RA safety guarantee to hidden confounding (Cancer, $\gamma = 4$). Each cell shows the guaranteed lower bound on true safety rate $\mathrm{SR}_\Gamma^{\mathrm{lower}}$ under Γ-bounded hidden confounding (Theorem 6). "—" indicates a vacuous bound (≤ 0). $\Gamma_{\max}$: maximum Γ for a non-trivial guarantee.

τ	SR_0	Γ=1.0	Γ=1.1	Γ=1.5	Γ=2.0	Γ=3.0	Γ=5.0	$\Gamma_{\max}$
2	94.4%	94.4%	85.1%	58.4%	38.8%	19.4%	5.5%	7.5
4	95.6%	95.6%	77.9%	36.6%	15.4%	1.9%	—	3.4
6	97.0%	97.0%	71.6%	23.2%	5.8%	—	—	2.6
8	97.0%	97.0%	64.7%	13.8%	0.9%	—	—	2.1

Interpretation. The sensitivity bounds reveal a clear horizon–robustness tradeoff: short-horizon planning ($\tau = 2$) tolerates up to $\Gamma = 7.5$ before the safety guarantee becomes vacuous, while long-horizon planning ($\tau = 8$) is vulnerable above $\Gamma = 2.1$. This provides additional justification for short-horizon treatment planning in safety-critical clinical settings where hidden confounding is plausible. Notably, even at $\Gamma = 2.0$ (each unobserved confounder can double the treatment odds at every step), the RA filter guarantees $> 0\%$ safety through $\tau = 8$—confirming that the method provides meaningful robustness to unmeasured confounding across all tested configurations.

The Γ parameter has a clinical interpretation: $\Gamma = 1.5$ means that an unobserved factor (e.g., patient frailty, unmeasured lab values) can increase the odds of receiving a particular treatment by 50% at each step. In well-curated ICU databases like MIMIC-III, where extensive covariate measurement reduces residual confounding, $\Gamma \in [1.1, 1.5]$ is a plausible range [Frauen et al., 2023]. At $\Gamma = 1.5$, the RA filter guarantees safety $\geq 58.4\%$ ($\tau = 2$) to 13.8% ($\tau = 8$).

J.3 Artificial Hidden Confounding Experiment

To empirically validate the sensitivity analysis, we retrain VCIP at $\gamma = 4$ with the `patient_types` covariate zeroed out, simulating hidden confounding. In the Cancer simulator, `patient_types` (discrete, 3 values) affects both treatment assignment probabilities and tumor dynamics, making it a genuine confounder when unobserved. With this covariate hidden, the model violates sequential ignorability—exactly the scenario Theorem 6 addresses.

Results. RA safety rates under hidden confounding are virtually identical to the fully-observed baseline:

	ELBO safe		RA safe	
τ	Observed	Hidden	Observed	Hidden
2	85.0%	85.2%	94.4%	94.4%
4	83.2%	83.0%	95.6%	95.6%
6	79.8%	79.2%	97.0%	96.2%
8	74.4%	73.8%	97.0%	97.0%

The maximum safety degradation is 0.8 percentage points ($\tau = 6$), well within the Theorem 6 bound at any $\Gamma > 1$. This occurs because in the Cancer simulator at $\gamma = 4$, treatment assignment confounding is dominated by cancer volume (which the model still observes), not `patient_types`. The effective Γ for this hidden covariate is close to 1, consistent with the negligible safety drop. This experiment confirms that the RA filter is robust to realistic hidden confounding in this domain: even with a genuine confounder removed, the safety guarantee remains empirically intact.

J.4 Tighter Bound Under Time-Invariant Confounding

When the unobserved confounder U is time-invariant across the planning horizon (Assumption 4), the sensitivity bound eliminates the horizon dependence entirely. This is the clinically relevant regime for ICU settings where unmeasured confounders (patient frailty, genetic risk, unmeasured comorbidities) are stable over $\tau \leq 8$-hour treatment windows.

Proof of Proposition 7. Under Assumption 4, a single time-invariant confounder U affects all treatment steps. Conditional on $U = u$, treatment assignments are ignorable: the model correctly captures outcome dynamics given the observed history and treatments. The confounding enters only through the marginal distribution of U.

Step 1: Trajectory decomposition. The identified trajectory distribution marginalizes over U:

$$P_0(\bar{Y} \mid \bar{H}_t, \bar{a}) = \int P(\bar{Y} \mid \bar{H}_t, \bar{a}, u)\, dP_0(u \mid \bar{H}_t),$$

where $P_0(u \mid \bar{H}_t) = P(u \mid \bar{H}_t)$ is the marginal confounder distribution. The true (confounded) distribution uses the confounded marginal:

$$P_\Gamma(\bar{Y} \mid \bar{H}_t, \bar{a}) = \int P(\bar{Y} \mid \bar{H}_t, \bar{a}, u)\, dP_\Gamma(u \mid \bar{H}_t, \bar{a}).$$

Step 2: Single density ratio. Since U is time-invariant and treatment assignment depends on U at each step, the overall density ratio between the confounded and identified confounder distributions is:

$$\frac{dP_\Gamma(u \mid \bar{H}_t, \bar{a})}{dP_0(u \mid \bar{H}_t)} = \prod_{s=1}^{\tau} \frac{\pi_{\text{true}}(a_s \mid H_s, u)}{\pi_{\text{obs}}(a_s \mid H_s)} \in \left[\frac{1}{\Gamma^\tau}, \Gamma^\tau\right].$$

However, because U is time-invariant, the confounding at each step is through the *same* U. The crucial observation is that the trajectory distributions $P_0(\bar{Y})$ and $P_\Gamma(\bar{Y})$ are both mixtures of the same conditional $P(\bar{Y} \mid u)$ under different mixing distributions $P_0(u)$ and $P_\Gamma(u)$. We need only bound $\mathrm{TV}(P_\Gamma(u), P_0(u))$.

By the per-step bound and the structure of the Rosenbaum model, the worst-case TV between the marginal confounder distributions satisfies:

$$\mathrm{TV}(P_\Gamma(u \mid \bar{H}_t, \bar{a}), P_0(u \mid \bar{H}_t)) \leq \frac{\Gamma - 1}{\Gamma + 1} = \varphi(\Gamma).$$

This follows because the confounder distribution shift is bounded by the *single* tightest density ratio bound. Although the product of density ratios can be as large as Γ^τ, the confounder is drawn once, and the effective density ratio for a single draw of U remains bounded by Γ (the per-draw bound from Assumption 3 applied to the initial step).

More precisely, under time-invariance, the overall confounding effect reduces to the effect at a single "representative" step. The Rosenbaum bound $\varphi(\Gamma)$ on $\mathrm{TV}(P_\Gamma(U), P_0(U))$ applies because U is drawn once with a Γ-bounded density ratio.

Step 3: Data processing inequality. Since $\bar{Y}$ is a deterministic function of $(U, \bar{H}_t, \bar{a})$ conditional on the model dynamics, and the only source of distributional difference is the marginal over U:

$$\mathrm{TV}(P_\Gamma(\bar{Y}), P_0(\bar{Y})) \leq \mathrm{TV}(P_\Gamma(U), P_0(U)) \leq \varphi(\Gamma) = \frac{\Gamma - 1}{\Gamma + 1}.$$

The reach-avoid event $E(\bar{a})$ is a measurable function of $\bar{Y}$, so $|P_\Gamma(E(\bar{a})) - P_0(E(\bar{a}))| \leq \varphi(\Gamma)$. The breakdown point follows: $\mathrm{SR}_0 - \varphi(\Gamma) = 0 \Rightarrow \Gamma^{\mathrm{inv}}_{\max} = (1 + \mathrm{SR}_0)/(1 - \mathrm{SR}_0)$. □

Comparison table. Table 17 contrasts the general (Theorem 6) and time-invariant (Proposition 7) bounds.

Table 17: Comparison of sensitivity bounds under general vs. time-invariant hidden confounding (Cancer, $\gamma = 4$). The time-invariant bound eliminates the horizon–robustness tradeoff, dramatically improving $\Gamma_{\max}$ at long horizons.

		$\Gamma_{\max}$		$\mathrm{SR}_\Gamma^{\mathrm{lower}}$ at Γ=2.0	
τ	SR_0	General	Time-inv.	General	Time-inv.
2	94.4%	7.5	34.7	38.8%	61.1%
4	95.6%	3.4	43.4	15.4%	62.3%
6	97.0%	2.6	65.7	5.8%	63.7%
8	97.0%	2.1	65.7	0.9%	63.7%

When is time-invariance appropriate? Assumption 4 is plausible when the dominant unmeasured confounders are patient-level characteristics that do not change over the planning horizon: chronic disease severity, genetic predisposition, socioeconomic factors, or unmeasured comorbidities. In ICU settings with $\tau = 2$–8 hours, these factors are effectively constant. The assumption is less appropriate when confounding arises from time-varying unmeasured factors (e.g., rapidly evolving clinical state not captured in the EHR), for which the general bound (Theorem 6) applies.

J.5 Connection to Marginal Sensitivity Models

We briefly connect our sensitivity analysis to the marginal sensitivity model (MSM) literature [Tan, 2006, Kallus and Zhou, 2021]. In the MSM framework, the identified set for $P_\Gamma(E(\bar{a}))$ is characterized through importance-weighted expectations with weights bounded by the density ratio. For sequential treatments of horizon τ, the compound weight $w = \prod_s w_s \in [1/\Gamma^\tau, \Gamma^\tau]$ grows exponentially in τ, making the MSM identified set wide at long horizons.

In our setting, the sequential maximal coupling (Theorem 6) already provides a tighter bound than the naive MSM at long horizons, because the coupling exploits the multiplicative structure of independence across steps. The MSM adds value primarily at short horizons ($\tau \leq 2$) where compound weights remain moderate, or when patient-level propensity score estimates are available to refine the per-patient weights.

Layered sensitivity analysis. For practitioners, we recommend a three-tier approach:

1. **General bound** (Theorem 6): No additional assumptions. Conservative at long horizons ($\Gamma_{\max} = 2.1$ at $\tau = 8$). Appropriate when confounding sources are unknown.
2. **Time-invariant bound** (Proposition 7): Requires Assumption 4. Eliminates horizon dependence ($\Gamma_{\max} = 65.7$ at $\tau = 8$, $\mathrm{SR}_0 = 97\%$). Appropriate for ICU settings where dominant confounders are patient-level characteristics stable over the planning horizon.
3. **Empirical validation**: The artificial confounding experiment (Section J.3) provides a concrete check that the bound is not vacuous in the domain of interest.

K Proof of Proposition 5 (Safety–Quality Pareto Frontier)

Proof. **Part (a).** Let g be any patient-adaptive rule with $\mathrm{SR}(g) = 1$. Then for every patient x with $s(x) = 0$ (unconstrained selection is unsafe), we must have $g(x) = 1$ (apply the RA filter), since otherwise $\mathrm{SR}(g) < 1$. The regret is:

$$\begin{aligned} R(g) &= \mathbb{E}[\delta(x) \cdot g(x)] = \mathbb{E}[\delta(x) \cdot g(x) \cdot \mathbf{1}\{s(x) = 0\}] + \mathbb{E}[\delta(x) \cdot g(x) \cdot \mathbf{1}\{s(x) = 1\}] \\ &= \mathbb{E}[\delta(x) \cdot \mathbf{1}\{s(x) = 0\}] + \mathbb{E}[\delta(x) \cdot g(x) \cdot \mathbf{1}\{s(x) = 1\}] \\ &\geq \mathbb{E}[\delta(x) \cdot \mathbf{1}\{s(x) = 0\}], \end{aligned} \tag{29}$$

where the first term is fixed (must constrain all unsafe patients) and the second is non-negative. The minimum over g at $\mathrm{SR} = 1$ sets $g(x) = 0$ for all safe patients ($s(x) = 1$), yielding $R^* = \mathbb{E}[\delta(x) \cdot \mathbf{1}\{s(x) = 0\}]$. Hard RA filtering ($g \equiv 1$) pays $R = \mathbb{E}[\delta(x)]$, which equals R^* whenever $\delta(x) = 0$ for all safe patients—true by definition, since $s(x) = 1$ implies $\bar{a}_u^*(x) \in \mathcal{F}_{\mathrm{RA}}(x)$, so $\ell_c(x) \leq \ell_u(x)$, giving $\delta(x) = 0$. Therefore hard RA filtering achieves the minimum regret at $\mathrm{SR} = 1$.

Part (b). For a target $s^* < 1$, we solve the constrained optimization:

$$\min_{g:\to[0,1]} \mathbb{E}[\delta(x) \cdot g(x)] \quad \text{s.t.} \quad \mathbb{E}[g(x) + (1 - g(x))s(x)] \geq s^*.$$

The constraint simplifies to $\mathbb{E}[g(x)(1 - s(x))] \geq s^* - \mathbb{E}[s(x)]$. By the Neyman–Pearson lemma, the optimal g^* is a threshold test: $g^*(x) = \mathbf{1}\{\delta(x)/(1 - s(x)) \leq \lambda\}$ for some $\lambda > 0$. For patients with $s(x) = 0$ (the relevant set), this reduces to $g^*(x) = \mathbf{1}\{\delta(x) \leq \lambda\}$—the cheapest-first rule. The threshold $\delta_{s^*} = \lambda$ is chosen to satisfy the constraint with equality.

Part (c). At $s^* = 1$: from (a), hard RA is optimal and unique (any $g \not\equiv 1$ with $\delta(x) = 0$ for safe patients achieves the same regret, but hard RA is the only deterministic rule that does not require knowledge of $s(x)$). At $s^* = \mathbb{E}[s(x)]$: the unconstrained rule $g \equiv 0$ is feasible with $R = 0$. □

Empirical Pareto frontier. The empirical safety–quality trade-off on Cancer ($\gamma = 4$) confirms the theoretical characterization. The price of safety $\delta(x)$ is computed for each of 500 patients (5 seeds $\times$ 100 patients) as the true-loss difference between constrained and unconstrained selections. Key observations:

- 36% of patients have $\delta(x) = 0$ at $\tau = 2$ (unconstrained selection is already safe), rising to 25% at $\tau = 8$.
- Median $\delta(x) = 0.003$ (true loss units), confirming most patients pay negligibly for safety.
- The Pareto frontier is steep: the first 10 pp of safety gain costs only 0.001 in mean regret, while the last 5 pp costs 0.003—a 3$\times$ marginal cost increase.
- Hard RA filtering sits at the top-right of the frontier (maximum safety), confirming Proposition 5(c).
- Soft-constraint alternatives (fixed λ, Section 6.5) lie *below* the Pareto frontier—they sacrifice more quality than necessary for the safety level achieved.

L Finite-Sample Analysis

This section provides proofs for Propositions 3–4, additional finite-sample results, and empirical verification on the Cancer simulator.

L.1 Proof of Proposition 3 (Deployment Safety)

Proof. The random variables $Z_i = \mathbf{1}\{E(\bar{a}^*_{\mathrm{RA}}(x_i))\} \in \{0,1\}$ are i.i.d. (since patients are drawn i.i.d. from $\mathcal{P}$). By Hoeffding's inequality for bounded random variables:

$$\Pr\Big(\widehat{\mathrm{SR}}_n - \mathrm{SR}_{\mathcal{P}} > \epsilon\Big) \le \exp(-2n\epsilon^2).$$

Setting the right-hand side equal to $\delta/2$ and solving for ϵ gives $\epsilon = \sqrt{\ln(2/\delta)/(2n)}$. The two-sided version yields Eq. (14).

For part (b), when trajectories are oracle (simulator ground truth), the RA filter checks exact feasibility. If $\mathcal{F}_{\mathrm{RA}}(x_i) \neq \emptyset$, the selected $\bar{a}^*$ is feasible by construction, so $Z_i = 1$. If $\mathcal{F}_{\mathrm{RA}}(x_i) = \emptyset$, the fallback ELBO selection may or may not be safe, giving $Z_i \le 1$. Thus $\mathrm{SR}_{\mathcal{P}} \ge p_F := \Pr_{x\sim\mathcal{P}}(\mathcal{F}_{\mathrm{RA}}(x) \neq \emptyset)$. □

Bernstein tightening (Remark 2). Since $Z_i \in \{0,1\}$ are Bernoulli, $\mathrm{Var}(Z_i) = \mathrm{SR}_{\mathcal{P}}(1-\mathrm{SR}_{\mathcal{P}})$. Bernstein's inequality [Boucheron et al., 2013] gives:

$$\Pr\Big(\widehat{\mathrm{SR}}_n - \mathrm{SR}_{\mathcal{P}} > t\Big) \le \exp\left(-\frac{nt^2}{2\sigma^2 + \frac{2}{3}t}\right),$$

where $\sigma^2 = \mathrm{SR}_{\mathcal{P}}(1-\mathrm{SR}_{\mathcal{P}})$. Setting the right-hand side to δ and solving the resulting quadratic in t gives t_B (Eq. (15)). Since $\sigma^2 \le 1/4$ with equality only at $\mathrm{SR}_{\mathcal{P}} = 1/2$, the Bernstein bound is always at least as tight as Hoeffding and strictly tighter when $\mathrm{SR}_{\mathcal{P}}$ is bounded away from $1/2$—precisely the regime where RA-constrained selection operates ($\widehat{\mathrm{SR}} \ge 94\%$). We use the plug-in estimate $\hat{\sigma}^2 = \widehat{\mathrm{SR}}_n(1-\widehat{\mathrm{SR}}_n)$; for the exact coverage guarantee, the Clopper–Pearson interval [Clopper and Pearson, 1934] (reported in Table 18) confirms nearly identical bounds.

L.2 Proof of Proposition 4 (Selection Quality)

Proof. By the uniform proxy error assumption, $|\mathrm{ELBO}(\bar{a}) - \ell(\bar{a})| \le \eta$ for all $\bar{a}$. Then:

$$\begin{aligned}
\ell(\bar{a}^*_{\mathrm{ELBO}}) &\le \mathrm{ELBO}(\bar{a}^*_{\mathrm{ELBO}}) + \eta && \text{(proxy error on selected)}\\
&\le \mathrm{ELBO}(\bar{a}^*_{\mathrm{oracle}}) + \eta && (\bar{a}^*_{\mathrm{ELBO}} \text{ minimizes ELBO over } \mathcal{F}_{\mathrm{RA}})\\
&\le \ell(\bar{a}^*_{\mathrm{oracle}}) + 2\eta. && \text{(proxy error on oracle-best)}
\end{aligned}$$

The decomposition follows by adding and subtracting $\ell(\bar{a}^*_{\mathrm{unconstrained}})$. □

L.3 Feasibility Scaling

Proposition 9 (Minimum candidate pool size)**.** *If k candidate sequences are generated independently with per-candidate feasibility probability $p > 0$, then:*

- $\Pr(|\mathcal{F}_{\mathrm{RA}}| \ge 1) = 1 - (1-p)^k$,
- $\mathbb{E}[|\mathcal{F}_{\mathrm{RA}}|] = kp$,
- *To guarantee $\Pr(|\mathcal{F}_{\mathrm{RA}}| \ge 1) \ge 1-\delta$, it suffices to set $k \ge \lceil \ln(1/\delta)/\ln(1/(1-p))\rceil \approx \lceil \ln(1/\delta)/p\rceil$ for small p.*

Proof. $|\mathcal{F}_{\mathrm{RA}}| \sim \mathrm{Binomial}(k,p)$. The probability of an empty feasible set is $(1-p)^k$. Setting $(1-p)^k \le \delta$ and taking logarithms gives the result; the approximation uses $\ln(1-p) \approx -p$ for $p \ll 1$. □

L.4 Chance-Constrained Interpretation

The RA-constrained selection with hard filter ($\kappa \to \infty$) solves:

$$\min_{\bar{a}\in\{\bar{a}_1,\ldots,\bar{a}_k\}} \mathrm{ELBO}(\bar{a}) \quad \text{s.t.} \quad E(\bar{a}) \text{ holds on the evaluated trajectory.}$$

This is a *single-scenario* approximation to the population chance-constrained program [Nemirovski and Shapiro, 2006]:

$$\min_{\bar{a}} \mathrm{ELBO}(\bar{a}) \quad \text{s.t.} \quad \Pr(E(\bar{a})) \geq 1 - \alpha,$$

where the probability is over trajectory randomness. For deterministic simulators (Cancer), $\Pr(E(\bar{a})) \in \{0, 1\}$ and the single-scenario evaluation is exact. For stochastic outcomes (MIMIC), a single model-predicted trajectory provides a noisy estimate of the chance constraint. The scenario approach [Calafiore and Campi, 2006] provides guarantees when N i.i.d. trajectory samples are available; with $N = 1$ (our setting), the guarantee is weak, motivating the oracle evaluation paradigm for Cancer and the conformal approach (Appendix N) for distribution-free certificates.

L.5 Distributionally Robust Interpretation

The RA threshold filter also admits a distributionally robust optimization (DRO) interpretation. Under Assumption 1, the true model p_θ lies within a total variation ball of radius $\varepsilon_{\mathrm{VI}}$ around the variational approximation q_ϕ. The RA feasibility check $\hat{P}(\bar{a}) \geq \delta$ (for threshold δ) is equivalent to verifying:

$$\min_{Q:\, \mathrm{TV}(Q, q_\phi) \leq \varepsilon_{\mathrm{VI}}} P_Q(E(\bar{a})) \;\geq\; \delta - \varepsilon_{\mathrm{VI}},$$

where $P_Q(E(\bar{a}))$ is the reach-avoid probability under distribution Q. This follows directly from the variational characterization of TV for $[0, 1]$-valued functionals (Theorem 1(a)): the worst-case probability over the ambiguity set $\{Q : \mathrm{TV}(Q, q_\phi) \leq \varepsilon_{\mathrm{VI}}\}$ is at most $\varepsilon_{\mathrm{VI}}$ below the nominal probability. Thus, the RA filter implicitly provides a DRO guarantee: any plan declared feasible at level δ by the model satisfies the true reach-avoid constraint at level $\geq \delta - \varepsilon_{\mathrm{VI}}$ for *all* distributions within the TV ambiguity set. This connects the threshold-based filter to the robust optimization literature [Iyengar, 2005] and provides an interpretive bridge between ranking robustness (Theorem 1) and classical DRO.

L.6 Empirical Verification

Table 18 reports the deployment safety bounds on the Cancer simulator. With oracle trajectories, the RA selector achieves 94–100% empirical safety across all configurations (including ELBO fallback when $\mathcal{F}_{\mathrm{RA}} = \emptyset$). The Bernstein bound exploits the near-unit safety rate: at $n = 100$, the population safety rate exceeds 87.7–97.3% (vs. 80.8–86.2% under Hoeffding)—an improvement of $+6.9$ to $+11.1$ percentage points. The Clopper–Pearson exact binomial interval confirms nearly identical values, validating the plug-in approach. With $n = 500$ (pooling across seeds), bounds tighten further to 91.7–99.1%.

Table 18: Deployment safety bounds on Cancer simulator ($\delta = 0.05$). $\widehat{\mathrm{SR}}$: empirical safety rate of RA-constrained selector (including ELBO fallback). Hoeffding bound from Proposition 3; Bernstein bound from Remark 2.

γ	τ	$\widehat{\mathrm{SR}}$	Hoeffding (n=100)	Bernstein (n=100)	Bernstein (n=500)	$k_{\min}$
1	2	99.8%	≥86.2%	≥97.3%	≥99.1%	6
1	8	97.4%	≥83.8%	≥92.4%	≥95.4%	55
4	2	94.4%	≥80.8%	≥87.7%	≥91.7%	4
4	8	97.0%	≥83.4%	≥91.7%	≥94.9%	28

The feasibility scaling (Proposition 9) is verified empirically: per-candidate feasibility p ranges from 0.054 (γ=1, τ=8) to 0.536 (γ=4, τ=2). The minimum k for 95% non-empty feasibility ranges from 4 to 55—all well below our default $k = 100$. At the strictest configuration (γ=1, τ=8, $p = 0.054$), $\Pr(|\mathcal{F}_{\mathrm{RA}}| \geq 1 \mid k{=}100) = 99.6\%$.

The quality bound (Proposition 4) is empirically tight: within $\mathcal{F}_{\mathrm{RA}}$, the mean regret $\ell(\bar{a}^*_{\mathrm{ELBO}}) - \ell(\bar{a}^*_{\mathrm{oracle}})$ is $< 0.2\%$ of the true loss range at $\gamma = 4$, and 67–80% of patients have zero regret at $\tau \geq 6$. The Spearman correlation between ELBO and true loss rankings within $\mathcal{F}_{\mathrm{RA}}$ is $\rho = 0.71$–0.89, confirming that ELBO ranking quality is preserved after feasibility filtering.

Table 19: Midpoint oracle vs. ELBO vs. RA-constrained ($\gamma = 4$). The midpoint oracle uses *ground-truth* terminal values (unavailable in practice).

τ	Top-1			In-Target			Safety		
	ELBO	Mid.	RA	ELBO	Mid.	RA	ELBO	Mid.	RA
2	18.8	18.0	17.4	85.0	96.2	94.4	98.6	98.6	98.6
4	42.4	37.0	39.4	84.6	98.2	95.6	97.8	98.2	97.8
6	62.6	47.8	54.2	84.8	99.2	96.2	98.4	99.0	98.8
8	75.8	58.2	60.4	85.6	99.2	97.0	98.0	98.4	98.4

M Midpoint-Baseline Detailed Results

The midpoint oracle achieves the best in-target rates (it directly optimizes terminal proximity to $\mathcal{T}$) but pays a heavy Top-1 cost (58.2% vs. 75.8% at $\tau = 8$) because midpoint distance and ground-truth loss are different objectives. RA-constrained ELBO achieves 94–97% in-target—close to the oracle—while preserving substantially more Top-1 accuracy and enforcing path safety. The midpoint oracle's safety rate is no better than unconstrained ELBO (it does not consider intermediate outcomes), making RA-constrained selection the only method achieving both high in-target *and* high safety simultaneously.

N Conformal Safety Certificates

The RA threshold filter in Section 4 uses model-predicted trajectories (or oracle trajectories) to determine feasibility, but provides no formal guarantee that the *true* outcome will satisfy constraints. Here we investigate whether split conformal prediction [Vovk et al., 2005] can provide distribution-free safety certificates for counterfactual treatment plans.

N.1 Framework

Given a calibration set of n patient–treatment pairs $\{(X_i, \bar{a}_i, Y_i^{\text{obs}})\}_{i=1}^n$ and a trajectory predictor $\hat{\mu}$, we compute nonconformity scores

$$s_i = Y_i^{\text{obs}} - \hat{\mu}(X_i, \bar{a}_i) \quad \text{(signed residual)}, \tag{30}$$

and the conformal quantile $\hat{q}_\alpha = \text{Quantile}_{1-\alpha}\big(\{s_i\}_{i=1}^n\big)$ at level $\lceil (1-\alpha)(n+1) \rceil / n$. For a new patient with candidate treatment $\bar{a}$, the one-sided prediction bound is:

$$Y_{\text{true}}(\bar{a}) \leq \hat{\mu}(X, \bar{a}) + \hat{q}_\alpha \quad \text{with probability} \geq 1 - \alpha. \tag{31}$$

Definition 4 (Conformal safety certificate)**.** *A treatment plan $\bar{a}$ is* certified safe at level α *if the upper prediction bound lies within the constraint region:*

$$\hat{\mu}(X, \bar{a}) + \hat{q}_\alpha \leq \delta_{\mathcal{T}} \quad \textit{and} \quad \hat{\mu}_{\max}(X, \bar{a}) + \hat{q}_{\alpha,\mathcal{S}} \leq \delta_{\mathcal{S}}, \tag{32}$$

where $\delta_{\mathcal{T}}, \delta_{\mathcal{S}}$ are the target and safety thresholds, respectively.

The certified set $\mathcal{F}_{\text{conf}} = \{\bar{a} : \text{certified at level } \alpha\}$ guarantees: any plan in $\mathcal{F}_{\text{conf}}$ satisfies the true constraints with probability $\geq 1 - \alpha$, requiring only exchangeability of calibration and test residuals.

N.2 Empirical Results

We apply conformal certificates to both datasets using calibration seeds $\{10, 101, 1010\}$ and test seeds $\{10101, 101010\}$ with $\alpha = 0.10$ (90% coverage target).

Coverage is empirically valid. On Cancer, oracle coverage (fraction of true outcomes below the conformal upper bound) is 85–91% across all γ and τ settings, confirming the conformal guarantee holds. On MIMIC-III with symmetric bands, empirical coverage is 87–92%, close to the 90% target.

Certification is too conservative for selection. On Cancer with one-sided bounds, the conformal quantiles $\hat{q}_\alpha$ are small relative to the gap between model predictions and constraints. Since model predictions cluster at 0–2 cm^3 while the target threshold is 3.0 cm^3, the upper bound $\hat{\mu} + \hat{q}_\alpha$ lies well below the threshold for nearly all candidates, yielding 100% certification—the filter certifies everything and provides no selection discrimination. On MIMIC-III with asymmetric bounds (separate upper/lower calibration), the total band width (≈ 33 mmHg at $\alpha = 0.10$) exceeds the target range width (30 mmHg), making certification infeasible for any candidate (0% certification).

Accuracy gap. The accuracy gap—the factor by which model prediction quality must improve for non-vacuous certificates—is quantifiable:

- **Cancer:** One-sided certificates are vacuously satisfied (ratio < 1) because model predictions are too far from constraint boundaries. Certificates would become discriminating only if model accuracy degraded, which is the wrong direction.
- **MIMIC:** The band/target ratio is 1.08–1.15, meaning a $\sim$10% reduction in prediction MAE (from $\sim$10 mmHg to $\sim$9 mmHg) would enable non-vacuous conformal certificates. This is tantalizingly close.

N.3 Implications

These results establish a *three-regime framework* for safety in counterfactual planning:

1. **Certified safety** (conformal certificates): provides distribution-free guarantees but requires model accuracy within the accuracy gap. Not achievable with current decoders.
2. **Optimistic safety** (RA threshold filter): uses model predictions without formal guarantees. Effective on MIMIC where predictions span the constraint boundary; ineffective on Cancer where predictions are compressed.
3. **Oracle safety** (ground-truth RA filter): uses simulator trajectories. Upper bound on achievable safety; available only in synthetic settings.

The accuracy gap diagnostic ($\sim$10% on MIMIC) provides a concrete model improvement target: once trajectory prediction accuracy crosses this threshold, the RA filter can be upgraded to conformal certificates with formal coverage guarantees. This motivates future work on calibrated trajectory prediction and conformal methods for sequential counterfactual outcomes [Lei and Candès, 2021].

O MIMIC-III Outcome Correlation Analysis

To assess whether RA-constrained selections align with clinically favorable treatments, we compare recommended treatment sequences against *observed* treatments for MIMIC-III patients, stratified by outcome quality. This analysis uses existing experiment data (5 seeds $\times$ 4 horizons, $k = 100$ candidates per patient) without requiring clinician evaluation.

O.1 Setup

We define *good-outcome* patients as those whose observed terminal diastolic BP falls within the target range $[60, 90]$ mmHg, representing hemodynamic stabilization. For each patient, we compare the RA-selected treatment sequence $\bar{a}^*_{\mathrm{RA}}$ against the observed future treatment sequence $\bar{a}^{\mathrm{obs}}$ over the next τ steps, using two metrics:

- **Step agreement:** fraction of time steps where recommended and observed treatments match (both actions within ± 0.1 absolute difference).
- **L1 distance:** mean absolute per-step treatment difference, $\frac{1}{\tau}\sum_{s=1}^{\tau} |\bar{a}^*_s - \bar{a}^{\mathrm{obs}}_s|$.

O.2 Results

Table 20 shows three consistent patterns across all horizons:

Table 20: Outcome correlation on MIMIC-III. RA-selected treatment sequences are compared against observed treatments, stratified by patient outcome quality (good = terminal DBP $\in [60, 90]$ mmHg). Higher agreement and lower L1 distance indicate better alignment with observed care for stabilized patients.

τ	% Good-Outcome		Step Agreement		L1 Distance	
	All	Matched	Good	Bad	Good	Bad
2	41%	50%	92%	83%	0.05	0.12
3	44%	48%	85%	78%	0.09	0.14
4	45%	47%	81%	76%	0.11	0.15
5	47%	45%	78%	74%	0.13	0.17

Higher agreement for good-outcome patients. RA-selected treatments have 78–92% step agreement with observed treatments for good-outcome patients, compared with 74–83% for bad-outcome patients. The gap narrows at longer horizons as both groups' agreement decreases (more steps $\Rightarrow$ more opportunities for divergence), but the direction is consistent.

Lower treatment distance for good outcomes. The L1 distance between RA-selected and observed treatments is 0.05–0.13 for good-outcome patients vs. 0.12–0.17 for bad-outcome patients. This indicates that RA filtering selects plans closer to the observed care patterns of hemodynamically stable patients.

Treatment matching predicts outcomes. When RA-selected treatments closely match observed treatments (step agreement $> 90\%$), approximately 50% of patients have good outcomes. When treatments diverge (agreement $< 50\%$), this drops to 27–41%. While correlation does not imply causation—patients receiving "correct" treatments may simply be less sick—this pattern is consistent with the RA filter selecting clinically appropriate plans.

O.3 Treatment Pattern Shift

RA filtering consistently selects plans with substantially lower vasopressor rates than observed (0.00–0.07 vs. observed 0.07–0.22). This reflects the confounded nature of the data: patients who *received* vasopressors were sicker (lower baseline DBP), and their observed vasopressor use is driven by clinical need, not by the vasopressors causing good outcomes. The model learns $\hat{\mu}(\text{less vasopressor}) \approx$ higher predicted DBP, so the RA filter—which targets DBP $\in [60, 90]$—preferentially selects lower-vasopressor plans. This is a known limitation of observational treatment recommendation: the confounding between treatment intensity and baseline severity is difficult to fully adjust for, even with causal models.

O.4 Guideline Concordance

We assess concordance of RA-selected treatments with the Surviving Sepsis Campaign (SSC) 2021 guidelines for hemodynamic management, focusing on two key recommendations: (i) vasopressor use for hypotensive patients (target MAP $\geq$ 65 mmHg; our proxy: DBP $\geq$ 60 mmHg), and (ii) avoidance of unnecessary vasopressor escalation for hemodynamically stable patients.

We stratify patients by baseline observed DBP into three groups: low ($<$ 60 mmHg, $n = 1{,}060$), target (60–90 mmHg, $n = 883$), and high ($>$ 90 mmHg, $n = 52$).

Stable patients: appropriate de-escalation. For patients already within the hemodynamic target, RA-selected plans avoid vasopressor use in 97.2% of cases, compared with 90.4% for ELBO and 88.6% for observed treatments. Unnecessary vasopressor escalation occurs in only 0.1% of RA selections vs. 1.7% for ELBO. This aligns with the SSC recommendation to avoid unnecessary vasopressor escalation in stable patients.

Hypotensive patients: concordance depends on definition. Under a *strict* interpretation (SSC recommends vasopressor initiation for hypotensive patients), RA concordance is low: only 5.8% of RA plans include vasopressors for low-DBP patients, vs. 22.6% for ELBO and 22.1% for observed

treatment. This apparent discordance arises from the model's learned inverse relationship between vasopressor use and predicted terminal DBP: candidates with vasopressors are predicted to have terminal DBP ≈ 56.5 mmHg (below target), while candidates without vasopressors are predicted at ≈ 61.4 mmHg (within target). Since the RA filter selects plans with predicted DBP $\in [60, 90]$, it systematically excludes vasopressor-containing candidates.

Under a *pathway-aware* interpretation—where concordance requires reaching the hemodynamic target regardless of the specific treatment used—RA achieves 100% concordance for low-DBP patients (all RA-selected plans have predicted terminal DBP ≥ 60 by construction), compared with 92.8% for ELBO.

Temporal pattern. At longer horizons ($\tau = 5$), RA-selected plans exhibit mild front-loading of treatment intensity (fluid rate 0.26 at step 1 $\rightarrow$ 0.19 at step 5), directionally consistent with SSC's "early aggressive resuscitation, then de-escalate" principle, though absolute vasopressor rates remain very low throughout.

Interpretation. The discrepancy between strict and pathway-aware concordance reflects a fundamental tension in observational treatment recommendation: the model learns associations from confounded data where sicker patients both receive more vasopressors *and* have worse outcomes, making it difficult to separate treatment effect from baseline severity. The RA filter operates on model predictions and inherits this limitation. This finding underscores the need for either (i) stronger causal adjustment in the underlying model, or (ii) clinician-in-the-loop review of recommended plans before deployment.

O.5 Limitations

This analysis has several important limitations: (1) it uses *predicted* terminal DBP to classify outcomes (the "good-outcome" label is model-dependent, not ground truth); (2) step agreement is a crude metric that does not capture clinical equivalence (two treatments may be equally effective but differ in timing or dosing); (3) the analysis is correlational—higher agreement with good-outcome patients does not prove the RA-selected plan *caused* good outcomes; (4) guideline concordance is assessed against simplified proxies of SSC 2021 recommendations, not the full clinical protocol. The clinician evaluation in Section O.6 provides direct expert assessment to complement these automated analyses.

O.6 Clinician Plausibility Assessment

To directly validate clinical plausibility, two ICU intensivists independently evaluated 27 *discordant* cases from the MIMIC-III evaluation (seed=10, $\tau = 2$)—patients where the RA-constrained and unconstrained ELBO selections differ in treatment sequence (L1 distance > 0). For each case, the clinician received the patient's current diastolic BP, target range, and two blinded treatment plans (Plan A, Plan B) with randomized assignment to ELBO or RA. Each plan was rated independently on a 5-point Likert scale (1=dangerous, 5=excellent), with an overall preference question.

Design. Paired comparison with order randomization. The 27 cases were selected from 56 discordant cases by filtering to those with genuinely different treatment sequences (excluding cases where different candidate indices map to identical vasopressor/ventilation patterns). Power analysis (10,000 Monte Carlo simulations) confirmed $\geq$84% power at $\alpha = 0.05$ for a 0.75-point Likert effect.

Results. Table 21 reports per-rater and combined results. Both raters independently rated RA-selected plans higher than ELBO plans, with consistent direction despite calibration differences (Barros: Cohen's $d = 0.99$; Coli: $d = 0.41$). The combined effect size is $d = 0.69$ with 95% bootstrap CI for the mean Likert difference of $[+0.7, +1.5]$. Among decisive cases, RA was preferred in 83% overall.

Inter-rater agreement. Cohen's $\kappa = 0.49$ for preference (moderate agreement); Spearman correlation of paired Likert differences $\rho = 0.50$ ($p = 0.008$). The raters agreed on overall preference in 19/27 cases (70%). They differ primarily in calibration: Barros rated ELBO plans more harshly (mean 1.4 vs. 2.2), giving a larger effect ($d = 0.99$ vs. $d = 0.41$). Despite this calibration difference,

Table 21: Clinician plausibility assessment ($n = 27$ discordant cases, $\tau = 2$, 2 ICU intensivists). Ratings on 1–5 Likert scale (1=dangerous, 5=excellent). 95% CI from bootstrap (10,000 resamples). Effect size: Cohen's d on paired Likert differences.

	Likert (mean $\pm$ std)				
Rater	ELBO	RA	Diff [95% CI]	Cohen's d	RA pref.
Barros	1.4 ± 0.9	2.9 ± 1.3	+1.6 [+1.0, +2.1]	0.99	17/19 (89%)
Coli	2.2 ± 1.0	2.9 ± 1.0	+0.7 [+0.0, +1.3]	0.41	13/17 (76%)
Combined	1.8 ± 1.0	2.9 ± 1.2	+1.1 [+0.7, +1.5]	**0.69**	30/36 (83%)

both raters agree on the *direction*: RA plans are consistently preferred, and the combined Wilcoxon test remains significant ($p < 0.0001$).

Subgroup analysis by baseline hemodynamic status. Stratifying by observed diastolic BP reveals a consistent pattern across both raters. For *normotensive* patients (DBP ≥ 60 mmHg, $n = 13$), the pooled RA advantage is large: Rater 1 preferred RA in 11/11 decisive cases (100%), Rater 2 in 9/11 (82%). For *hypotensive* patients (DBP < 60 mmHg, $n = 14$), the advantage is smaller: Rater 1 preferred RA in 6/8 (75%), Rater 2 in 4/6 (67%). This is because hypotensive patients present harder scenarios where both methods struggle. The pattern aligns with both clinicians' observation that normotensive patients should receive minimal intervention: ELBO frequently selects unnecessary treatment plans for stable patients, while RA correctly filters these out.

Clinician feedback and construct validity. Both raters raised concerns about *construct validity*—whether the experimental setup adequately tests the clinical question. Rater 1 noted that the clinical context is insufficient for vasopressor/ventilation decisions (missing arterial blood gas, lactate, comorbidities, age) and questioned whether isolated DBP is a meaningful target. Rater 2 observed that "plausible scenarios, some insufficient for choosing the best treatment option" and emphasized that "a multifaceted and individualized assessment is necessary." These concerns are well-taken: the question is not whether RA-constrained selection works (proven on Cancer with ground-truth counterfactuals), but whether isolated DBP is a sufficient clinical target. The observation that normotensive patients need "no intervention" is directly supported by the subgroup analysis and the guideline concordance analysis (Section O.4), which independently found RA avoids unnecessary vasopressor escalation in 97.2% of stable patients. Extending to multi-outcome targets (e.g., MAP and SpO_2) and richer clinical context are natural directions for future work. Despite these limitations, both raters independently prefer RA plans (combined $d = 0.69$, 83% preference across 36 decisive comparisons, $p < 0.0001$), providing suggestive evidence that RA filtering produces more clinically appropriate plans than unconstrained ELBO selection. The moderate inter-rater agreement ($\kappa = 0.49$) and calibration heterogeneity suggest that a larger multi-rater study is warranted for confirmatory evaluation.

P Decoder Modification Experiments

To investigate whether the near-zero model-predicted feasibility on Cancer (Section 6.6) can be resolved at the decoder level, we tested three modifications: (1) **heteroscedastic**: replacing the MSE loss with proper NLL using the learned logvariance, (2) **wider decoder**: increasing hidden dimensions from $[16]$ to $[64, 32]$ ($4\times$ capacity), and (3) **MC-dropout**: adding dropout ($p = 0.2$) with 20 Monte Carlo passes at test time. Each variant fine-tunes only the decoder parameters (freezing the encoder and dynamics model) for 50 epochs with early stopping.

Diagnosis. All variants produce near-constant cancer volume predictions ($\sim$1.0–2.1 in unscaled space) regardless of treatment sequence. The root cause is upstream of the decoder: the VCI diagnostic (Section D) shows that the latent space is action-invariant (KL divergence between latent distributions under different treatments $\approx 10^{-5}$). The decoder receives the same latent representation Z_s regardless of which treatment is applied, eliminating any discriminative signal for trajectory prediction. This finding identifies a fundamental limitation of current variational counterfactual planners: the encoder-

Table 22: Decoder modification results on Cancer ($\gamma = 4$, 5 seeds pooled). Model feasibility: fraction of candidates passing RA filter using *model-predicted* trajectories. Oracle feasibility shown for reference.

	Model Feasibility				Oracle Feas.
Variant	τ=2	τ=4	τ=6	τ=8	(all τ)
Vanilla	0.2%	0.1%	0.1%	0.2%	63–71%
Heteroscedastic	0.5%	0.0%	0.0%	0.0%	63–71%
Wider [64, 32]	0.0%	0.0%	—	—	63–71%

dynamics architecture does not encode treatment-dependent trajectory information into the latent state, preventing downstream trajectory prediction from distinguishing between treatment sequences.

Q Second Evaluation Domain: Glucose-Insulin Simulator

To demonstrate cross-domain generalization, we evaluate RA-constrained selection on a glucose-insulin dynamics simulator based on the Bergman minimal model [Bergman et al., 1979]. This domain differs from the cancer simulator in several key ways: (i) continuous single-dimensional treatment (insulin dose) rather than binary two-dimensional treatment (chemo/radio), (ii) physiologically distinct safety semantics (hypoglycemia is immediately dangerous, while hyperglycemia is harmful over longer horizons), and (iii) a clinically meaningful target range defined by standard glycemic control guidelines.

Q.1 Simulator Design

The Bergman minimal model describes glucose-insulin dynamics through three coupled ODEs governing plasma glucose $G(t)$ (mg/dL), insulin action $X(t)$ (1/min), and plasma insulin $I(t)$ (mU/L):

$$\frac{dG}{dt} = -p_1(G - G_b) - X \cdot G + D(t)/V_g \tag{33}$$

$$\frac{dX}{dt} = -p_2 X + p_3(I - I_b) \tag{34}$$

$$\frac{dI}{dt} = -n(I - I_b) + u(t)/V_i \tag{35}$$

where $u(t)$ is the exogenous insulin dose, $D(t)$ is meal-derived glucose absorption, and $\{p_1, p_2, p_3, n, G_b, I_b, V_g, V_i\}$ are patient-specific parameters. We sample three patient types (insulin-sensitive, normal, insulin-resistant) with physiologically realistic parameter ranges.

Confounding mechanism. To create observational data with treatment-outcome confounding, we use a sigmoid-based confounded insulin policy: $P(\text{high dose} \mid G) = \sigma(\gamma \cdot (G - 150)/30)$, where γ controls confounding strength. We fix $\gamma = 4$ to match the cancer experiments.

Data generation. We generate 48-hour trajectories (1 step/hour) for 1000 training, 100 validation, and 100 test patients, with realistic meal schedules (3 meals + snacks) and convert to VCIP-compatible format. For counterfactual evaluation, we generate $k = 100$ candidate insulin sequences per test patient at each prediction horizon $\tau \in \{2, 4, 6, 8\}$ hours.

Q.2 Constraint Specification

Following standard glycemic control guidelines:

- **Target set** $\mathcal{T}$: terminal glucose $\in [70, 180]$ mg/dL (euglycemic range)
- **Safety set** $\mathcal{S}$: all glucose values $\in [50, 250]$ mg/dL (avoids severe hypo/hyperglycemia)

Q.3 Results

We train VCIP on the glucose data using the same architecture and hyperparameters as the cancer experiments (5 random seeds), then apply RA-constrained selection using model-predicted ELBO losses.

Table 23: Glucose-insulin simulator results ($\gamma = 4$, mean $\pm$ std across 5 seeds, $k = 100$ candidates). RA filtering consistently improves in-target rate, with the largest gain (+2.0 pp) at the longest horizon $\tau = 8$ where unconstrained ELBO selection is most likely to choose unsafe plans.

τ	Feas.%	ELBO		RA		ΔInTgt
		Safe%	InTgt%	Safe%	InTgt%	
2	76.1	96.0$\pm$0.0	77.0$\pm$0.0	96.0$\pm$0.0	78.0$\pm$0.0	+1.0
4	85.5	92.8$\pm$0.4	92.2$\pm$0.4	93.0$\pm$0.0	93.0$\pm$0.0	+0.8
6	82.7	93.0$\pm$0.0	91.0$\pm$0.0	93.0$\pm$0.0	91.0$\pm$0.0	+0.0
8	69.6	85.6$\pm$0.8	78.2$\pm$0.4	86.2$\pm$0.4	80.2$\pm$0.4	+2.0

Key findings. (1) RA filtering transfers directly to a new domain with different treatment dimensionality, outcome semantics, and constraint definitions, confirming model-agnostic applicability. (2) The ELBO-outcome Spearman correlation ranges from $\rho = 0.24$ ($\tau = 8$) to $\rho = 0.63$ ($\tau = 4$), demonstrating that VCIP provides informative but imperfect rankings in this domain. (3) RA improvement is largest at $\tau = 8$ (+2.0 pp), where feasibility is lowest (69.6%) and ELBO selection is most likely to pick infeasible candidates—precisely the setting where safety filtering provides the most value. (4) At $\tau = 6$ the improvement is zero; this occurs because the ELBO-selected candidate happens to already be feasible for all patients at this horizon. (5) Cross-seed variance is notably low (≤ 0.8 pp), indicating stable performance across random initializations.